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Graph Clustering using Self Organizing Maps.

AGOURZAM Abdelhadi

Master en Sciences Informatiques

`abdelhadi.agourzam@student.umons.ac.be`

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Mentor : Gauvain Devilez

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Chapter 1

Introduction

1.1 Motivation

Graphs describe objects whose meaning lies in their connections rather than in a list of measurements. A molecule is a set of atoms and the bonds between them. Program is a set of statements and the control that passes between them. Social network is a set of people and the relationships that join them. In each case not only the entities have information but the full structure.

Clustering such objects (grouping them so that inner cluster members are more similar than inter cluster members) is a natural analytical task. It is how one asks whether a collection of molecules falls into families, whether programs share architectural patterns, or whether networks come in recognisable types. Clustering is also the standard first step when the aim is exploratory, to discover what kinds of objects are present rather than to assign objects to true cluster labels.

Every clustering method rests on a notion of similarity. For data represented as vectors, two vectors of the same length can be compared coordinate by coordinate, and the Euclidean distance between them is so natural a choice that it is rarely presented as a choice at all. The coordinates are aligned by construction: the third number in one vector means the same thing as the third number in another.

Graphs offer no such alignment. Two graphs may have different numbers of nodes, and even when they have the same number there is no canonical way to decide which node in the first corresponds to which node in the second. A graph is defined only up to relabelling of its nodes, so any comparison must either search over the possible correspondences or avoid them entirely. Searching is expensive, deciding whether two graphs are isomorphic has no known polynomial algorithm, and measuring how far apart two non isomorphic graphs are is harder still. Avoiding the correspondence means summarising each graph by quantities that do not depend on how its nodes are named, and then comparing the summaries.

This is the decision the present work takes as its subject. Before any clustering algorithm can be applied to a set of graphs, a distance must be chosen, and the available choices are not variations on a single idea. They rest on different notions of what makes two graphs alike.

One family counts the work required to turn one graph into another. Under the graph edit distance (1; 2), similarity means that few insertions, deletions and re-bellings separate the two objects. A second family compares the eigenvalues of a matrix derived from the graph, typically the Laplacian. Here similarity means that the two graphs vibrate alike, in the sense that many global structural quantities are encoded in the spectrum. A third family compares iterated descriptions of local neighbourhoods, as in the Weisfeiler Lehman scheme, so that similarity means the two graphs contain the same local patterns in the same proportions. A fourth compares the distribution of shortest path lengths, making similarity a matter of how the graph is traversed rather than how it is wired.

These are not four approximations to one underlying quantity. There are four different questions about a graph, and there is no reason to expect their answers to agree. Two graphs may require few edits to interconvert while having quite different spectra or two graphs may have identical path length distributions while differing sharply in their local neighbourhoods.

Each distance also discards information, and necessarily so. A summary that did not discard anything would be equivalent to the graph itself and would return us to the correspondence problem it was introduced to avoid. Compression is the point. But compression is selective, what one summary keeps, another throws away ? A distance built on path lengths retains information about how spread out a graph is and loses information about which particular edges are present. A distance built on the spectrum retains certain global counts and loses others.

The consequence is that the choice of distance is not a preliminary technicality to be settled by convention or convenience. It determines which properties of a graph remain visible to everything downstream. A property that a distance does not encode is invisible to every algorithm that consumes that distance, no matter how the algorithm is configured or how much data it is given. The clustering method can only work with what the distance preserves.

This is easy to state and easy to overlook, because in practice the distance is usually selected the way a hyperparameter is selected, several are tried, and the one producing the best clustering score is kept. That procedure treats the distance as a tuning knob whose value is justified by the outcome it produces. The following sections argue that the procedure cannot deliver what it appears to, and that the choice deserves to be studied directly.

1.2 Clustering graphs with self organising maps

Once a distance is available, many clustering algorithms become applicable. Any method that operates on a matrix of pairwise distances rather than on coordinates can be used: k medoids, hierarchical agglomeration, spectral clustering. This work takes the self organising map as its primary method.

A self organising map (3) , in its original formulation, is a set of prototypes arranged on a fixed low dimensional grid, usually two dimensional. Each prototype occupies a position on the grid and holds a representative of the data. Training proceeds by repeatedly presenting an object, identifying the prototype closest to it (best

matching unit) and moving that prototype toward the object. Crucially, the prototype's neighbours on the grid are moved as well, by an amount that decreases with grid distance. The strength and reach of this neighbourhood influence shrink over training, so early passes organise the map globally and later passes refine it locally.

The neighbourhood update is what distinguishes the method. Without it, the prototypes would be adjusted independently and the algorithm would reduce to something like k means / k medoids. With it, prototypes that sit near one another on the grid are pulled toward similar regions of the data, so the final grid becomes an ordered arrangement where adjacent positions hold similar objects, and distant positions hold dissimilar ones. The map is therefore not only a partition but a layout, a two dimensional surface onto which the data has been projected while preserving neighbourhood relations.

The standard formulation assumes that prototypes can be moved continuously, which requires the objects to live in a vector space where a weighted average is defined. Graphs do not. There is no general way to average a set of graphs, like we do with numbers/vectors like what is the mean of this set of graphs ?

Two adaptations are available. The first is to construct prototypes in graph space directly, moving a prototype toward an input by applying a small number of edit operations that we call (weighted mean graph). This is closest in spirit to the original algorithm and was proposed in the framework this work follows (4). It requires an edit based distance and is computationally demanding, since each update involves searching over all possible edits.

The second is to constrain each prototype to be an element of the dataset. A prototype is then an index rather than a constructed object, and the update step selects, for each grid position, the dataset element minimising a neighbourhood-weighted sum of distances to all inputs. This is the median self organising map. It requires only a distance matrix, applies to any distance whatever, and reduces the update to a minimal index over the data. Its cost is expressiveness, a prototype must be a graph that actually occurs in the dataset, so the map cannot represent a region of graph space that the data does not populate (5).

The median formulation is adopted here. It is the only variant that can be applied unchanged to all four distances under study, which is a requirement for comparing them. An algorithm available for one distance and not another would confound the comparison at the outset.

The choice also carries a consequence that becomes relevant later. Because prototypes are indices, two grid positions can select the same dataset element. Should that occur, the neighbourhood weighted update computes identical quantities for both positions and returns identical results, so the two prototypes remain merged for the remainder of training. The map can therefore lose capacity during training, where two coinciding prototypes would generally separate again under subsequent updates.

What the map offers, relative to a plain partition, is the layout. A partition tells you which objects belong together. A map tells you which groups sit next to which other groups, and how sharply they are divided. If the training has succeeded, a class of objects occupies a connected region of the grid rather than being scattered across it, and the boundaries between regions correspond to genuine discontinuities

in the data. These are properties of the arrangement, and they are inspectable, one can visualise the map, examine which prototypes lie on boundaries, and see how the classes are laid out relative to one another.

This suggests a division of labour that this work adopts. A self organising map should be assessed on two questions rather than one. The first is whether its partition recovers the structure present in the data, which is the question asked of any clustering method and on which the map competes with the alternatives. The second is whether the arrangement it produces is meaningful, whether the spatial organisation reflects the data's organisation. Only the second is specific to the method, and it is the second that determines whether the map is worth its additional machinery. Both questions are addressed in what follows, and they are kept apart. Conflating them is a recurring source of confusion in the literature, where diagnostics designed to assess the quality of the map are sometimes used to judge the quality of the partition, or to select the map's parameters. Whether those diagnostics support such use is one of the questions this work examines.

1.3 Problem : analyse clustering results with different distance functions

Suppose the question is which of several graph distances is best suited to a clustering task. The natural experiment is to run the same algorithm with each distance on the same labelled dataset and compare the results. Whichever distance yields the highest agreement is declared the most appropriate.

This does not answer the question. The score is produced jointly by three things, and the experiment cannot separate them. The first is what the distance encodes ? If the property separating the classes is not represented in the distance at all, no score above chance is possible. The second is how far apart the classes happen to lie under that distance, encoding a property is not the same as separating classes by it, and a distance may respond so weakly that the difference between classes is small compared with the variation within them. The third is what the algorithm does with whatever separation exists. A single score is the product of all three and moves when any one changes. A low score might mean the distance is blind, or that the dataset was built so the classes barely separate, or that the structure was there and the algorithm missed it. These are different conclusions, and the experiment cannot distinguish them. The difficulty runs deeper, because the confound originates in the experimenter's own choices. A labelled graph dataset is built by partitioning graphs according to some property. But graph properties are correlated, graphs with more edges also tend to have more triangles, higher maximum degree, and smaller diameter. Selecting on one property selects, to a degree, on all of them. When a distance then separates the classes, it is unclear which property it responded to, because the alternatives were never separated within the dataset. The obvious remedy is to build a better dataset, holding the correlated properties fixed while varying the target. This is the right instinct and it partially works, but it does not fully succeed. Constraints accumulate faster than graphs satisfying them, and some controls are impossible in principle. More fundamentally, any dataset is built by selecting graphs according to a rule, and

the rule is itself a graph property. Controlling for the obvious confounds shifts the question to whatever the selection rule happens to track. The bias relocates rather than disappearing. The problem, then, is not carelessness. It is that the standard experimental form can not in principle attribute a result to the distance, because the dataset and the algorithm are entangled with it in the measurement. The response taken here is to measure the three factors separately. What a distance encodes can be established without any dataset and without any clustering, by examining the distance against the population of graphs it might be applied to. How far apart classes lie can be measured before an algorithm runs. What an algorithm achieves can be assessed on data whose separation is known rather than assumed. Designed datasets then regain a role, but a different one, not as evidence of what a distance can do, but as tests of predictions made from the independent measurements. A dataset built to demonstrate something proves little, since its designer knew what they wanted to see. A dataset used to check a prediction formed elsewhere is a good test, and its biases are tolerable because the prediction did not come from it. This restructuring determines the order of what follows. The characterisation of the distances comes first and uses no clustering. The geometry of the datasets comes second and uses no algorithm comparison. The clustering experiments come last, read against what the earlier stages established.

1.4 Research Questions

RQ1. What can each distance represent, and what is it unable to distinguish? This concerns the distance alone. Because every distance summarises a graph before comparing, some pairs of non-isomorphic graphs necessarily receive identical summaries and therefore a distance of zero. Identifying these establishes a hard ceiling on everything downstream, independent of any dataset or algorithm. Beyond outright indistinguishability, the question extends to what the non-zero values are composed of: whether a distance is essentially a restatement of some elementary graph statistic, or whether it carries structure that no standard description captures.

RQ2. What determines whether a clustering recovers a property? Given that a distance encodes a property, recovery is not guaranteed. This question asks what quantity governs it. Candidates include how strongly the distance responds to the property, how far apart the classes end up, and how the geometry of the resulting distance matrix is shaped. Answering it yields a calibration, a way of saying, before any clustering is run, what recovery should be expected, which in turn converts every later result from an isolated number into a deviation from a prediction.

RQ3. Is the choice of distance a hyperparameter or a modelling decision? A hyperparameter is a setting whose value is justified by the outcome it produces, and which can therefore be selected by trying alternatives. A modelling decision is a commitment about what the analysis treats as meaningful, and is justified by argument rather than by outcome. The distinction is testable: if the distance behaves as a hyperparameter, the variation in recovery attributable to it should be comparable to the variation attributable to other tunable choices, such as the clustering algorithm.

If it behaves as a modelling decision, it should dominate them, and the property being sought should determine which distance is appropriate rather than the other way around.

RQ4. What does a self organising map contribute beyond a partition? If the map's partition performs comparably to that of simpler methods, its value must lie in the arrangement. This question asks whether the arrangement is meaningful whether classes occupy connected regions of the grid, whether the layout is better organised than a random one, and whether properties of the arrangement carry information about the quality of the clustering. Certain metrics are used to judge whether a SOM is good. This experiment asks, do those metrics work in our experiments? And more specifically, can they tell you how many neurons to put on the grid?

RQ5. Does the distances characterisation can be transferred to a property that no distance was designed for? In the first four experiments, we picked the properties ourselves. That means we might have rigged the game without realising it. So for the final test, we picked something we had no control over (chromatic number, a famously hard problem (6)). We wanted to know if does our framework still work when we step outside the lab?

1.5 Contributions

C1. A framework for characterising a graph distance independently of any clustering task. We show that a distance can be described directly, what it is in effect a function of, and how it responds when a single graph property is changed under otherwise identical conditions by using only the population of graphs, with no labelled dataset and no algorithm involved. This turns the choice of distance from something inferred from downstream scores into something measured, and it yields characterisations of the four distances studied here that are sharper and in places contrary to what their definitions suggest.

C2. A demonstration that recovery is governed by class separation rather than by algorithm choice. By generating clustering problems of deliberately varied difficulty and measuring that difficulty rather than assuming it, we compare SOM against k medoids and hierarchical clustering on equal terms. The comparison establishes what proportion of clustering performance is attributable to the distance, to the algorithm, and to how the dataset was constructed, and identifies a separation threshold below which no method recovers structure reliably.

C3. An account of what a SOM offers beyond a partition, and of why its size cannot be chosen from its own diagnostics. We evaluate the map as a spatial arrangement rather than only as a clustering, and show that its layout carries information about clustering quality that the partition alone does not. We then examine the quantity conventionally used to select the map's size and find that,

although it behaves exactly as the selection procedure assumes, it is unrelated to the quality of the resulting clustering.

C4. A way of using biased datasets as evidence. Any dataset built by selecting graphs on a property is biased: the classes differ in that property and in everything correlated with it, and no amount of care in the design removes this. Our response is to never use such a dataset to establish what a distance can do, only to check a prediction made elsewhere. We characterise the distances on graphs drawn at random, with no classes and no labels, then write a function that turns those characterisations into a predicted outcome for every dataset we built. The function is fixed before any clustering is run and is reported in full, so its reasoning can be checked. The predictions were correct in more than half of time and, more importantly, never inverted, no case predicted to fail succeeded, and none predicted to succeed failed.

Chapter 2

Background and Related Work

This chapter establishes the mathematical and algorithmic foundations on which the rest of the thesis is built. Every object, distance, algorithm, and evaluation metric used in the experiments is defined here and tied to its implementation in the code. The full code and other documents are presented in our github account "<https://github.com/oscarRickovic/Project>"

2.1 Graph Theoretic Preliminaries

A *graph* is a pair $G = (V, E)$ where V is a finite set of nodes and $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$ is a set of edges. The number of nodes is $n = |V|$ and the number of edges is $m = |E|$. All graphs in this thesis are undirected and simple.

A *node label* is a map $\ell : V \rightarrow \Sigma$. In the code labels are stored as the node attribute `label`; the alphabet is $\Sigma = \{A, B\}$.

The *degree* of a node v is the number of edges incident to it, denoted $\deg(v)$. The *maximum degree* is

$$\deg_{\max}(G) = \max_{v \in V} \deg(v),$$

and the *minimum degree* is

$$\deg_{\min}(G) = \min_{v \in V} \deg(v).$$

The *degree variance* is

$$\text{Var}_{\deg}(G) = \frac{1}{n} \sum_{v \in V} (\deg(v) - \bar{d})^2, \quad (2.1)$$

where $\bar{d} = \frac{1}{n} \sum_{v \in V} \deg(v)$ is the average degree.

The *number of connected components* is $\kappa(G)$. A graph is *connected* if $\kappa(G) = 1$.

The *diameter* of a connected graph is

$$\text{diam}(G) = \max_{u, v \in V} d_G(u, v), \quad (2.2)$$

where $d_G(u, v)$ is the length of the shortest path between u and v . If G is disconnected, $\text{diam}(G)$ is defined as the diameter of its largest connected component.

A *triangle* is a set of three nodes $\{u, v, w\}$ such that all three edges (u, v) , (v, w) , and (u, w) belong to E . For a node v , let $\tau(v)$ denote the number of triangles containing v . The total number of triangles in the graph is

$$t(G) = \frac{1}{3} \sum_{v \in V} \tau(v). \quad (2.3)$$

The *local clustering coefficient* of a node v with $\deg(v) \geq 2$ is

$$c(v) = \frac{2e(v)}{\deg(v)(\deg(v) - 1)}, \quad (2.4)$$

where $e(v)$ is the number of edges between the neighbours of v . If $\deg(v) < 2$, we set $c(v) = 0$. The *average clustering coefficient* is

$$C(G) = \frac{1}{n} \sum_{v \in V} c(v). \quad (2.5)$$

The *clique number* $\omega(G)$ is the size of the largest complete subgraph of G .

The *chromatic number* $\chi(G)$ is the smallest integer k such that the nodes of G can be coloured with k colours and no adjacent nodes share a colour. In the code $\chi(G)$ is computed by exact backtracking search; this is feasible because all experimental graphs have at most $n = 9$ nodes.

Two graphs G and H are *isomorphic* (7) if there exists a bijection between their node sets that preserves adjacency and node labels. The *Weisfeiler Lehman (WL) graph hash* is a polynomial time proxy for isomorphism testing (8). It iteratively refines node colours by hashing the multiset of neighbours' colours. After h iterations the multiset of colours is hashed to a canonical string. In the code this is `nx.weisfeiler_lehman_graph_hash` (9). Graphs with different hashes are certainly non isomorphic, for $n \leq 9$ hash collisions are negligible in practice.

The *adjacency matrix* of G is A , and the *degree matrix* is

$$D = \text{diag}(\deg(v_1), \dots, \deg(v_n)).$$

The *Laplacian* is

$$L = D - A. \quad (2.6)$$

The eigenvalues of L are real and non-negative:

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n.$$

The spectrum is invariant under graph isomorphism.

The *nauty* (10; 11) suite is used to generate exhaustive catalogues of small graphs. The program `geng` emits graphs in *graph6* byte-string format, which is parsed by `networkx.from_graph6_bytes`. Command-line flags control connectedness (`-c`) and edge count ranges. Catalogues are cached in memory to avoid regeneration, which dominates the runtime for $n \leq 9$.

2.2 Graph Distance Measures

Four distances are studied. They are presented from the most exact and expensive to the most approximate and cheap. Each takes two graphs and returns a non-negative number.

2.2.1 Graph edit distance

The graph edit distance (1; 2) is the cost of the cheapest sequence of operations turning one graph into the other. The operations are inserting and deleting nodes and edges, and substituting one node label for another. Each operation has a cost; here every operation costs 1, which is the standard unit-cost model.

$$d_{\text{GED}}(G, H) = \min_{s \in \mathcal{S}(G, H)} \sum_{o \in s} c(o),$$

where $\mathcal{S}(G, H)$ is the set of edit sequences transforming G into H and $c(o)$ is the cost of operation o .

GED is zero exactly when the two graphs are isomorphic, so unlike the other three it never confuses distinct graphs. That completeness is why it is used here as a reference. The price is cost: computing it exactly is NP hard, since the search ranges over node correspondences. In practice a time budget is imposed and the best value found within it is returned, which is an upper bound on the true distance rather than the distance itself. The size of that budget is therefore a parameter of the method and is reported with the results (12).

Under unit costs, GED counts operations (1; 2). Two edits of equal cost are equally far apart regardless of what they do to the graph, a point Chapter 3 examines directly.

2.2.2 Spectral distance

The spectral distance (13) compares the Laplacian spectra defined in Section 2.1. Each graph is summarised by its sorted eigenvalues, and the distance is the Euclidean distance between the two vectors:

$$d_{\text{Spec}}(G, H) = \left(\sum_{i=1}^N (\lambda_i(G) - \lambda_i(H))^2 \right)^{1/2}.$$

When the graphs have different orders the shorter spectrum is padded with zeros up to $N = \max(n_G, n_H)$, which is consistent with treating the smaller graph as having isolated extra nodes.

Computing a spectrum costs $O(n^3)$, so the distance is cheap at the sizes used here. It ignores node labels entirely: two graphs with the same structure and different labels are at distance zero. Because the spectrum is not a complete invariant, cospectral graphs are also at distance zero (14).

2.2.3 Weisfeiler Leman distance

The WL distance compares the colour histograms produced by the refinement of Section 2.1. After h iterations each graph yields a vector $\phi(G)$ counting how often each colour occurs. The WL kernel is the inner product $k(G, H) = \langle \phi(G), \phi(H) \rangle$, which is large when two graphs share many local patterns.

A kernel is turned into a distance in two steps. It is first normalised,

$$\hat{k}(G, H) = \frac{k(G, H)}{\sqrt{k(G, G)k(H, H)}} \in [-1, 1],$$

which removes the effect of graph size, and then converted by

$$d(G, H) = \sqrt{2 - 2\hat{k}(G, H)},$$

the Euclidean distance between the normalised feature vectors. The same conversion is used for the shortest path distance below. A distance obtained this way is Euclidean by construction, since it is a genuine distance between vectors.

WL reads node labels, since they seed the initial colouring. It inherits the incompleteness of the refinement: graphs the test cannot separate are at distance zero.

2.2.4 shortest path distance

The shortest path distance summarises a graph by how it is traversed rather than how it is wired. For each graph, count how many node pairs are at distance 1, at distance 2, and so on, giving a histogram $\psi(G)$. The kernel is $k(G, H) = \langle \psi(G), \psi(H) \rangle$, converted to a distance by the same normalisation as above.

Computing all pairwise shortest paths costs $O(n^3)$ by Floyd Warshall, or less on sparse graphs. Node labels are ignored. The summary is severe: a graph of order n is reduced to at most $n - 1$ numbers, so many graphs necessarily share a histogram.

2.2.5 What each distance ignores

Because each distance is built on an incomplete invariant, each is blind to something by construction. This is a consequence of the definitions and not an experimental result, Chapter 3 measures how often it happens.

Distance	Reads node labels	Zero for
GED	yes	isomorphic graphs only
Spectral	no	cospectral graphs
WL	yes	graphs the WL test cannot separate
SP	no	graphs with equal path-length histograms

2.3 Clustering Algorithms

All three algorithms below take a precomputed distance matrix rather than coordinates, which is what allows the same algorithm to be run with each of the four distances. Keeping the algorithm fixed while the distance varies is a requirement for comparing distances at all.

2.3.1 k -medoids

kmedoids (15) partitions the data into k groups, each represented by one of the data points, called its medoid. It alternates two steps until nothing changes: assign every point to the nearest medoid, then replace each medoid by the point of its group minimising the total distance to the rest of the group.

It differs from k -means in that a centre must be an actual data point. That is essential here, since no average of a set of graphs is available.

2.3.2 Agglomerative hierarchical clustering

Agglomerative clustering (16) starts with every object in its own cluster and repeatedly merges the two closest clusters until the desired number remains. What “closest” means is set by the linkage rule. Two are used here: *average linkage*, where the distance between clusters is the mean distance between their members, and *complete linkage*, where it is the largest such distance. Average linkage tends to produce balanced clusters, complete linkage compact ones.

The method is deterministic: it has no random initialisation and returns the same answer every time.

2.3.3 self organising maps

A self organising map (3). is a set of prototypes placed on a fixed grid, usually two dimensional. Each prototype holds a representative of the data and occupies a position on the grid. Training repeats three steps: present an object, find the prototype closest to it (the *best matching unit*), and move that prototype and its grid neighbours toward the object. The influence on a neighbour falls off with grid distance, following a Gaussian whose width shrinks as training proceeds, so early passes organise the map globally and later ones refine it locally.

The neighbourhood update is what separates a SOM from k -medoids. Because prototypes near one another on the grid are pulled toward similar data, the final grid becomes an ordered layout: adjacent positions hold similar objects. The result is therefore both a partition and a two dimensional arrangement.

2.3.4 The median self organising map

The classical update moves a prototype continuously toward an object, which requires a weighted average. Graphs have none. Two adaptations exist. One constructs

prototypes in graph space by applying edit operations, which needs an edit-based distance and is expensive. The other, used here, constrains each prototype to be an element of the dataset (5).

Write D for the $n \times n$ distance matrix and p_j for the index of the prototype at grid position j . Let b_i be the best matching unit of object i and let $H_{jk} = \exp(-\|g_j - g_k\|^2/2\sigma^2)$ be the neighbourhood weight between grid positions j and k , where g_j is the grid coordinate of position j . Each epoch performs

$$b_i = \arg \min_j D_{i,p_j}, \quad p_j = \arg \min_q \sum_i H_{j,b_i} D_{i,q},$$

so every prototype moves to the dataset element minimising a neighbourhood-weighted sum of distances. The width σ decays geometrically over the epochs.

This formulation is used throughout for three reasons. It needs only a distance matrix, so it applies unchanged to all four distances. It is deterministic given a seed, with no convergence difficulties. And its prototypes are real graphs, which can be inspected directly.

A prototype here is an index into the dataset, so two grid positions can hold the same graph. This is common. In the early epochs the neighbourhood is wide, so every neuron computes its update over nearly the whole dataset and most of them select the same graph. The neighbourhood narrows later and the prototypes partly separate, but not completely, a neuron sharing its graph with another is never the closest prototype for anything, so it has no data of its own to pull it away. Left uncorrected, (eg : a grid of twenty five positions finished our runs holding about six distinct graphs).

The implementation repairs this after every update. Each graph is scored by its distance to the nearest prototype, so a high score marks a graph the map represents badly. Any neuron holding a graph that another neuron already holds is then moved to the worst scoring graph not currently held by anyone. Since a graph is at zero distance from itself, the moved neuron immediately becomes the closest prototype for at least one graph. On the datasets used here this gives the full requested number of distinct prototypes in every run.

2.4 Evaluation Metrics

Metrics are split by what they need. External metrics compare a partition against known labels. Internal metrics use only the distance matrix and the partition. The distinction matters, an internal metric can say whether a partition is compact, but never whether it is the partition that was wanted.

2.4.1 External metrics

Adjusted Rand index. The Rand index counts pairs of objects placed together or apart in both partitions, divided by the total number of pairs. The adjusted form

subtracts the value expected by chance:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}.$$

It is 1 for identical partitions, 0 for chance, and negative when agreement is worse than chance. Because it counts pairs, splitting a correct class into several smaller ones is penalised even though nothing is misassigned. Wherever labels exist, ARI is the measure of recovery used here (17).

Separation AUC. This measures whether the labels are visible in the distance matrix at all, before any clustering. Take every pair of objects, label it 1 if the two come from different classes and 0 otherwise, and use the distance as a score for predicting that label. The area under the resulting ROC curve is the separation AUC. It equals the probability that a randomly chosen between-class pair is further apart than a randomly chosen within-class pair. A value of 0.5 means the distance carries no information about the classes (18; 19).

2.4.2 Internal metrics

Silhouette. For each object, let a be its mean distance to the other members of its own cluster and b its mean distance to the members of the nearest other cluster. The silhouette of that object is $(b - a) / \max(a, b)$, and the score is the mean over all objects. It lies in $[-1, 1]$, with higher values meaning tighter, better separated clusters (20).

The score is computed with the same distance that produced the partition. It can therefore say that a partition is compact under that distance, but it cannot compare one distance against another, since each is judged by its own standard.

Quantization error. The mean distance from each object to its best matching unit. It measures how well the prototypes summarise the data and decreases as prototypes are added (5).

Topographic error. The fraction of objects whose closest and second-closest prototypes are not neighbours on the grid. It measures whether the layout preserves neighbourhood relations. A raw value is difficult to interpret on its own, so it is reported against the value obtained when the grid positions are randomly permuted; a ratio near 1 means the layout is no better arranged than chance (21).

U-matrix. A visualisation rather than a score. Each grid position is shaded by the average distance from its prototype to the prototypes at neighbouring positions. Light regions mark boundaries between dissimilar areas of the map, so cluster structure can be read off the picture directly (5).

2.5 Why Controlled Datasets Are Needed

Chapter 3 builds datasets in an unusual way: classes are constructed, then forced to have identical edge counts, and chance levels are obtained by shuffling labels rather than looked up in a table. This section explains why that is necessary and names the ideas involved.

The problem: graph properties travel together. Suppose we want to know whether a distance can detect diameter. The obvious test is to build three classes of graphs with diameters 2, 4 and 6, and see whether the distance separates them.

The trouble is that a graph with a small diameter usually also has more edges, a higher maximum degree and more triangles than a graph with a large diameter. These properties come as a package. So if the distance separates the classes, we cannot say what it responded to. It might be diameter, or it might be edge count, which happens to differ between the classes because we selected on diameter.

A quantity that varies alongside the one we care about, and could explain the result just as well, is called a *confounder*. Edge count is a confounder for diameter, and for almost every other graph property.

First remedy: matching. We can remove a confounder by forcing it to be identical across classes. Group the available graphs by edge count, then build each class using the same number of graphs from each group. Afterwards every class has the same distribution of edge counts, so edge count can no longer explain any difference between them. This is called *matching* (22).

Matching is exact and easy to check: one simply compares the average edge count per class and confirms it is the same. Its drawback is that it throws graphs away. Requiring equal edge counts discards graphs with no partner in the other classes, and requiring equal edge counts *and* equal maximum degree discards many more. Add enough requirements and no graphs survive.

Some requirements cannot be met at all. A connected graph has no isolated node, so its minimum degree is at least 1; a graph with three components on nine nodes usually has one. Matching minimum degree across those two classes is therefore impossible, not merely difficult. Chapter 3 reports where this occurs.

Second remedy: statistical control. When matching is not possible, a confounder can be removed by arithmetic instead of by selection. Take every pair of graphs and record three things, the distance between them, how much they differ in edge count, and how much they differ in the property of interest. Then fit a single equation across all the pairs at once, expressing the distance as a sum of contributions from each property. The contribution attributed to the property of interest is what remains after edge count has already been credited with everything it can explain. No graphs are discarded, and the method keeps working when matching has run out of pairs. The price is an assumption, the equation supposes that each property contributes independently and that the contributions add together. Matching needs no

such supposition, because if two classes have identical edge counts then edge count cannot explain anything about the difference between them.

Chapter 3 uses both. Matching builds the labelled datasets, where the classes must be made comparable. Regression characterises the distances on graphs drawn at random, which were never grouped into classes at all.

Knowing what counts as chance. Separation AUC is 0.5 when a distance carries no information about the classes. That is the value expected with infinitely many graphs. With forty five graphs, and some of them structurally similar, a distance can reach 0.53 or 0.54 by luck alone.

To find the real chance level, we shuffle the class labels at random, recompute the AUC, and repeat a few hundred times. This gives the range of values the statistic takes when the labels mean nothing. A result is treated as real only if it exceeds the upper end of that range. This is a *permutation test*, and it is used here instead of the theoretical 0.5 because the theoretical value is optimistic at these sample sizes (23).

How large is the difference? A test says whether a difference exists. It does not say whether the difference is large enough to matter. For that we use an *effect size* (24) : the difference between two group means divided by their pooled standard deviation. It answers how many standard deviations apart are they?, which is independent of the units involved.

This matters because the four distances produce numbers of completely different magnitude. GED returns small integers, the shortest path distance returns values near zero. An effect size can be compared across all of them, a raw difference cannot.

2.6 Related Work

This section places the components of Chapter 3 in context. Each paragraph concerns one part of the experimental pipeline, how graphs are compared, how they are clustered, how a self organising map is adapted to them, and how the results are validated.

Comparing graphs. A collection of graphs can be clustered in two ways. One turns each graph into a vector of features and applies ordinary clustering to the vectors, which fixes the feature set in advance. The other keeps the graphs as graphs, defines a distance between them, and applies any method that accepts a distance matrix. Chapter 3 takes the second route, so the four distances of Section 2.2 are the point at which modelling assumptions enter (8; 25).

Comparisons between such distances are typically made through a downstream task, a classifier or clustering method is run with each distance on some collection, and the one giving the highest score is preferred. A score obtained this way depends on the distance, the dataset and the algorithm jointly, so it does not by itself say what the distance encodes. Blocks 2 to 4 of Chapter 3 measure the distances directly instead, without any clustering.

Clustering on a distance matrix. Several standard algorithms need only pairwise distances. Hierarchical agglomeration merges the closest clusters repeatedly, with average and complete linkage giving different notions of closeness (16). k -medoids (15) partitions the data around representatives that are themselves data points. Both are used in Chapter 3 as baselines against which the self organising map is compared, and both return a partition and nothing further.

Self organising maps for graphs. Existing work on graph SOMs generally fixes a single distance and studies the resulting map, so how the map would behave under a different notion of similarity is left open. Chapter 3 varies the distance while keeping the map's formulation fixed, and examines both directions of the interaction: what the map can recover depends on what the distance encodes, and what the map displays (which regions of the grid lie beside which, and how sharply they divide) depends on it as well. Block 9 therefore studies the map as an arrangement in its own right, and not only as a source of partitions.

Validating a clustering. When labels are available, agreement measures such as the adjusted Rand index compare a partition against them. When labels are not available, internal indices are used instead, and silhouette and quantization error are commonly employed to select the number of clusters or the size of a map. Two limitations of this practice are already recognised: an index computed inside one representation cannot compare two representations, and an index measuring how compactly the data has been summarised need not track whether the clustering is correct. Block 9 tests the selection procedures built on these indices directly, by asking whether the map size they choose is the size that in fact gives the best recovery (20; 5).

Chapter 3

Experiments and Results

3.1 Protocol

This section states everything shared by the experiments that follow, how the levels of analysis are separated, how graphs are generated, how the datasets are built, which metrics are introduced by this work, and how significance is assessed.

3.1.1 Three levels of analysis

Chapter 1 argued that a clustering score confounds three things. The experiments keep them apart by assigning every measurement to one of three levels.

- **L1, the distance.** What the distance encodes, measured on the distance matrix or on the distance function itself. No dataset with classes and no clustering algorithm is involved.
- **L2, the geometry.** How far apart the classes of a given dataset lie under a given distance. Involves a labelled dataset but no algorithm.
- **L3, the partition and the map.** What an algorithm produces, and how well it matches the labels when labels exist.

The rule applied throughout is that *a metric may only answer questions at its own level*. Silhouette is an L3 quantity computed inside one distance, so it cannot be used to compare distances. Quantization error describes how well prototypes summarise data, so it cannot be used to judge a partition. Where labels exist, the adjusted Rand index is the measure of recovery and no internal index overrides it.

3.1.2 Graph generation

All graphs are produced by exhaustive enumeration rather than by sampling from a random graph model. For a given order, and optionally a range of edge counts and a connectivity constraint, every non isomorphic graph satisfying the constraints is generated with **geng** from the **nauty** package (10; 11).

Orders five to nine are used. The counts of connected graphs are 21, 112, 853, 11,117 and 261,080 respectively, so enumeration remains practical throughout this range and becomes impractical beyond it. This bounds every result in the chapter to small graphs, which is stated again in the limitations.

Two properties of exhaustive pools matter. A quantity computed over a complete set is a property of the population and needs no confidence interval, which is what makes the results of Section 3.2 exact rather than estimated. And when a dataset must match several covariates across its classes, drawing from a complete pool makes the matching as feasible as it can be, if no suitable graphs exist in the complete set then none exist at all, so an infeasible design reflects a fact about graphs rather than an unlucky sample.

3.1.3 Building labelled datasets

A labelled dataset is built by choosing a target property, partitioning the range of that property into classes, and drawing graphs from the corresponding parts of an exhaustive pool. As explained in Section 2.5, this also separates the classes on every property correlated with the target.

The response is a *confound ladder*: the same target property is used to build several datasets with increasing amounts of control.

- **R1** controls nothing beyond the number of nodes. Classes differ in the target and in whatever accompanies it.
- **R2** additionally matches edge count exactly across classes.
- **R3** additionally matches degree statistics, which vary by dataset and are stated where each is defined.

Matching is exact and stratified. Graphs are grouped by the tuple of covariates being matched, only groups present in every class are used, and each class draws the same number of graphs from each group. Afterwards the matched covariates have identical distributions in every class and cannot explain any separation that remains. The cost is that graphs without a partner are discarded, so the pool shrinks with each covariate added, and some combinations become infeasible. Where a rung could not be built it is reported as infeasible rather than replaced by a weaker one.

Verification is by direct inspection. For each dataset the mean of each covariate is computed per class, and the standard deviation of those class means is reported. A value of zero means the covariate is matched.

3.1.4 The dataset families

Five families are used, in four roles.

edit-wide (positive control). Order nine, three classes defined by edge count: 8–9, 17–18 and 26–27 edges. The target property is edge count itself, so no ladder is possible: matching edge count would destroy the classes. Its purpose is to confirm that the pipeline detects a property that every distance should see. A distance failing here is not working.

conn-wide (test). Order nine, three classes defined by the number of connected components: one, two and three. Component count is exactly the multiplicity of the eigenvalue zero in the Laplacian, so it is encoded in the spectral representation by construction. R2 matches edge count, R3 additionally matches maximum and minimum degree.

geom-wide (test). Order nine, three classes defined by diameter: two, four and six. The diameters are spaced widely on purpose. R2 matches edge count, R3 additionally matches minimum degree. Matching maximum degree as well proved infeasible at this order and is reported as such.

pure-label (blindness control). Order seven, three classes defined by the proportion of nodes carrying a label B : none, half, all. The three classes are built on the *same* graph structures, so every structural covariate is identical across classes by construction and only the labelling differs. A distance that ignores node labels is therefore provably unable to separate the classes, and its measured separation gives a direct reading of the empirical chance level.

colorability (case study). Three classes defined by chromatic number. R1 fixes order nine and exactly fifteen edges but leaves clique number free. R2 uses Mycielski’s construction (26; 27), which raises the chromatic number while keeping the graph triangle free. R3 fixes order eight and matches edge count exactly, contrasting $\chi = 2$ against $\chi = 3$.

Each dataset is generated with four independent data seeds, giving four draws from the same design. Table 3.1 reports the resulting roster.

3.1.5 Metrics introduced in this work

The metrics of Section 2.4 are standard. Four further quantities are used here and are defined now, since they do not appear in the literature in this form.

Separation, d . The effect size between the within class and between-class distance distributions. Writing $\bar{d}_{\text{between}}$ and $\bar{d}_{\text{within}}$ for their means and s_p for their pooled standard deviation,

$$d = \frac{\bar{d}_{\text{between}} - \bar{d}_{\text{within}}}{s_p}.$$

It answers “how many standard deviations apart are the two kinds of pair?” and is scale-free, so it is comparable across distances whose values differ by orders of magnitude. It is used as the L2 measure of how far apart the classes lie (24).

dataset	round	graphs	classes	mean $ E $	$ E $ spread	δ spread
edit-wide	R1	45	3	17.5	7.10	1.21
conn-wide	R1	45	3	11.9	1.00	0.52
conn-wide	R2	45	3	11.1	0.00	0.48
conn-wide	R3	45	3	8.8	0.00	0.00
geom-wide	R1	45	3	11.9	1.39	0.22
geom-wide	R2	45	3	11.6	0.00	0.00
geom-wide	R3	45	3	11.6	0.00	0.00
pure-label	R1	36	3	10.8	0.00	0.00
colorability	R1	60	3	15.0	0.00	0.48
colorability	R2	33	3	34.9	15.29	0.21
colorability	R3	50	2	9.6	0.00	0.04

Table 3.1: Dataset roster. The last two columns give the standard deviation of the per-class mean edge count and the per class mean minimum degree. A value of 0.00 means the covariate is matched across classes and can no longer explain any separation. **edit-wide** retains a large edge count spread by design, since edge count is its target. **colorability** R2 shows a large spread because triangle free construction changes the size of the graph, a point returned to in Section 3.10.

Topographic error ratio. A raw topographic error is hard to read, since its scale depends on the grid and on the data. The ratio divides the observed error by the error obtained when the grid coordinates are randomly permuted, averaged over forty permutations:

$$\text{TE ratio} = \frac{\text{TE}(\text{observed layout})}{\text{TE}(\text{shuffled layout})}.$$

A ratio of 1 means the arrangement is no better organised than a random one, values below 1 mean it is better. This makes the quantity interpretable without reference to any external standard.

Territory pieces and concentration. Each neuron is assigned the majority true class of the graphs mapped to it, giving each class a *territory* on the grid. For a class, *pieces* is the number of grid-connected components its territory splits into, and *concentration* is the fraction of the territory lying in its largest component (Of all the graphs that truly belong to this class, what fraction landed in the class’s largest contiguous patch on the grid?). A class occupying one connected region has *pieces* = 1 and *concentration* = 1. Both are reported as means over classes. They measure the map as an arrangement rather than as a partition, and have no counterpart for algorithms that return only a partition.

Covariate baseline. For each elementary graph property c , a distance matrix is built from that property alone, $D_{ij}^{(c)} = |c_i - c_j|$, and its separation AUC computed. The covariate baseline is the largest such value over all properties, excluding the target property itself where one is defined. It answers: could a single elementary number have separated these classes as well as the distance did? A distance not exceeding this baseline has demonstrated correlation with the classes but not structural sensitivity.

3.1.6 Seeds, chance levels and baselines

Seeding. Two sources of randomness are separated. The *data seed* controls which graphs are drawn from the pool, four are used per dataset. The *algorithm seed* controls initialisation, between three and five are used depending on the experiment. Results are averaged over both unless stated otherwise.

Permutation nulls. The asymptotic chance level for separation AUC is 0.5, but with 45 graphs and structurally similar members the empirical chance level is higher. For every measured AUC the class labels are shuffled three hundred times (23), the statistic recomputed, and the 95th percentile of the resulting distribution reported. A cell is declared *blind* only when its observed AUC falls at or below that percentile. Observed nulls in this chapter lie between 0.52 and 0.55.

Verdicts. Combining the null and the covariate baseline gives three categories, applied mechanically:

- **blind** when AUC at or below the permutation null;
- **succeeds** when AUC above the covariate baseline,
- **confounded** when above the null but not above the baseline.

3.1.7 Implementation

The map. The median self organising map of Section 2.3 is used throughout, with 40 epochs (5) and a neighbourhood width decaying geometrically from $\max(\text{rows}, \text{cols})/2$ to 0.3.

Merging. A map with K neurons produces K groups, which cannot be compared against k true classes directly, since the adjusted Rand index penalises splitting even when nothing is misassigned. Unless stated otherwise, prototypes are merged by average-linkage agglomeration (16) down to the true number of classes before ARI is computed, so that every algorithm compared produces the same number of clusters. The unmerged value is reported separately where over-segmentation is of interest.

Computation. Distance matrices are computed once and cached on disk, keyed by the graph set and the distance. GED dominates the cost, at order nine with 45 graphs a single matrix takes roughly few minutes, against under a second for the other three. For this reason GED is computed on one data seed per dataset while the other distances use all four.

Software. Graphs are handled with `networkx`, numerical work with `numpy` and `scipy`, clustering baselines with `scikit-learn`, and graph generation with `nauty` (10; 11) . The complete notebook is provided as supplementary material and every table and figure in this chapter is produced by it (28; 29; 30; 31).

3.2 Experiment 1: What Each Distance Cannot Distinguish

3.2.1 Question

Every distance in this study compares graphs through a summary rather than through the graphs themselves. Whenever two different graphs produce the same summary, the distance between them is exactly zero, to that distance they are the same object. No clustering algorithm, grid size or random seed can ever separate such a pair.

This experiment asks how often that happens. For each distance, what proportion of graphs cannot be told apart from at least one other graph, and how large do the groups of mutually indistinguishable graphs become?

The answer is a ceiling on everything that follows. If a distance merges a large share of the graph space, every later result obtained with it is bounded by that fact, and the bound belongs to the distance rather than to any dataset or method.

3.2.2 Design

The measurement is arithmetic rather than statistical. Each distance is built on an invariant, and two graphs sharing that invariant are at distance zero (14). The invariants are:

- **Spectral** : the Laplacian spectrum. Two graphs with the same spectrum are cospectral and receive a distance of zero.
- **WL** : the colour multisets produced by $h = 3$ iterations of Weisfeiler Leman refinement. Graphs the refinement cannot separate produce identical colour histograms and hence identical feature vectors.
- **SP** : the histogram of shortest path lengths. Graphs with the same histogram produce the same feature vector.
- **GED** : no invariant. Exact graph edit distance with positive costs is zero only between isomorphic graphs, so it merges nothing.

3.2.3 Data

All connected graphs at orders five, six, seven and eight, generated exhaustively with **nauty**: 21, 112, 853 and 11,117 graphs respectively. No classes, no labels, no sampling. Order nine is not included because the exact characteristic polynomial computation over 261,080 graphs was outside the compute budget, the trend across the four orders is reported instead.

n	distance	distinct values	collapsed	fraction	largest group
5	Spectral	21	0	0.000	1
5	WL	21	0	0.000	1
5	SP	11	15	0.714	5
5	GED	21	0	0.000	1
6	Spectral	110	4	0.036	2
6	WL	109	6	0.054	2
6	SP	28	100	0.893	16
6	GED	112	0	0.000	1
7	Spectral	793	115	0.135	3
7	WL	832	42	0.049	2
7	SP	74	826	0.968	84
7	GED	853	0	0.000	1
8	Spectral	10251	1611	0.145	6
8	WL	10749	675	0.061	8
8	SP	198	11070	0.996	840
8	GED	11117	0	0.000	1

Table 3.2: Indistinguishability over the complete set of connected graphs at each order. “Distinct values” is the number of different summaries the distance can produce, “collapsed” counts graphs sharing a summary with at least one other graph. GED is included for reference: exact edit distance is zero only between isomorphic graphs.

3.2.4 Procedure

For each order, every graph is processed once and three keys are computed: the integer characteristic polynomial of its Laplacian, its Weisfeiler Leman hash at three iterations, and its sorted shortest path length histogram. Graphs are grouped by each key independently. For each distance we then report the number of distinct keys, the number of graphs sharing a key with at least one other graph, and the size of the largest group.

3.2.5 Results

The shortest path distance is the most degenerate by a wide margin, and becomes more so with size. At order five it already merges 71% of graphs. At order eight it maps 11,117 distinct graphs onto 198 different values, with 99.6% of graphs having at least one twin and a largest group of 840 mutually indistinguishable graphs. Two graphs drawn at random from this population receive a distance of exactly zero roughly four percent of the time.

The spectral distance collapses nothing at order five, then 3.6%, 13.5% and 14.5% at orders six, seven and eight. Its largest group grows from two to six. Cospectral graphs are not exceptional at these sizes.

The Weisfeiler Leman distance is the least degenerate of the three, with a collapse fraction between 4.9% and 6.1% across orders six to eight and no clear upward trend. Until order eight all of its groups contain exactly two graphs.

Figure 3.1 shows both quantities against order.

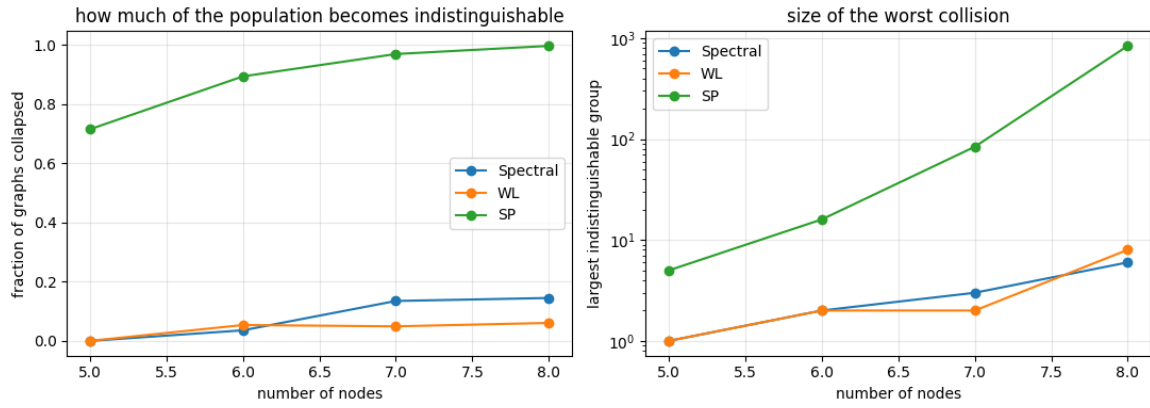


Figure 3.1: Left: fraction of the complete graph set that shares its summary with at least one other graph. Right: size of the largest indistinguishable group, on a logarithmic scale. SP grows steeply in both, WL is flat.

3.2.6 Interpretation

Three statements follow, each holding for connected graphs at orders five to eight and for the specific parameterisations used here.

The shortest path distance discards most of the graph. At order eight it distinguishes 198 classes of graph where 11,117 exist. This is not a defect of an implementation but a consequence of the summary: a graph of order n is reduced to at most $n - 1$ counts, and there are far more graphs than distinct histograms. Any result obtained with SP anywhere in this thesis is bounded by it, and any SP clustering of a broad sample of small graphs is operating on data that is largely degenerate.

The three tractable distances differ substantially in how much they retain. Ordering them by collapse fraction at order eight gives WL at 6.1%, Spectral at 14.5% and SP at 99.6%. This is a property of the distances themselves, established without reference to any task, and it is the first sense in which the choice between them is consequential rather than incidental.

The trends differ in direction. Between orders six and eight, SP rises from 89% to 99.6% and its largest group from 16 to 840, while WL stays near 5–6% with a largest group of 8. Spectral rises but far more slowly. Within the range tested, SP degrades with graph size and WL does not. Whether these trends continue beyond order eight is not established here, larger orders were not enumerated.

Two limits should be stated alongside these results. The measurement concerns *exact* indistinguishability only, it says nothing about whether the non-zero distances a summary produces are informative, which is the subject of the next two experiments. And the GED row reflects exact edit distance, whereas the approximation actually computed in later experiments uses a time budget and returns an upper bound, so it may fail to distinguish graphs that exact GED would.

3.3 Experiment 2: What Each Distance Is a Function Of

3.3.1 Question

Experiment 1 established what each distance cannot separate at all. Most pairs of graphs, however, are not identical under any of these distances: they receive some non-zero number. This experiment asks what that number is made of.

When the spectral distance reports that two graphs are 3.2 apart, what is it responding to? Is it that one has more edges, a longer diameter, a denser core? Or is it responding to something that no elementary description of a graph captures?

The question matters in two ways. It distinguishes a distance that is essentially a restatement of some simple graph statistic from one that carries additional structure. And, because the measurement uses no designed dataset, its results can be used to predict behaviour on designed datasets without inheriting their bias, which is what Experiment 6 does.

3.3.2 Design

The design choice that makes this experiment useful is that *there is no dataset*. Graphs are drawn uniformly from the complete population, there are no classes, no labels and no property being separated. Since nothing was designed, nothing can be biased by the design.

Every pair of graphs contributes one observation: the distance between them, and the absolute difference between them in each of seven elementary properties. A linear model is then fitted with the distance as the response and the seven differences as predictors, and three quantities are read from it.

- **Marginal variance** of a property: the R^2 obtained when that property is the only predictor. What it explains on its own.
- **Unique variance** of a property: the drop in R^2 when that property is removed from the full model. What it explains that no other property could.
- R^2_{all} : what all seven explain together. Its complement, $1 - R^2_{\text{all}}$, is the share of the distance that no elementary property accounts for.

The difference between marginal and unique variance measures confounding. If a property explains a great deal alone but little uniquely, most of its apparent influence is shared with other properties that are themselves driven by it.

This is the statistical control described in Section 2.5. Unlike matching, it keeps every observation and remains available at any graph size, at the cost of assuming a linear form for the relationship.

distance	graphs	R_{all}^2	unexpl.	$ E $	Δ	δ	tri	clust	diam
GED	60	0.421	0.579	0.059	0.000	0.003	0.001	0.004	0.000
Spectral	200	0.954	0.046	0.169	0.000	0.000	0.007	0.000	0.000
WL	200	0.412	0.588	0.046	0.002	0.004	0.005	0.001	0.004
SP	200	0.796	0.204	0.219	0.000	0.005	0.007	0.002	0.019

Table 3.3: Unique variance: what each property explains that no other property can. Degree variance is omitted from the table for width and is 0.064, 0.013, 0.060 and 0.001 respectively. Bootstrap 95% confidence intervals on all entries are at most 0.019, the widest being GED’s edge count and degree-variance terms.

distance	$ E $	Δ	δ	triangles	clustering	diameter	deg. var.
GED	0.330	0.082	0.092	0.277	0.110	0.047	0.088
Spectral	0.926	0.183	0.328	0.701	0.260	0.188	0.022
WL	0.304	0.102	0.159	0.274	0.082	0.035	0.094
SP	0.751	0.182	0.167	0.396	0.216	0.301	0.002

Table 3.4: Marginal variance: what each property explains on its own. Comparing with Table 3.3 shows how much of each property’s apparent influence is shared with the others.

3.3.3 Data

Two hundred connected graphs of order seven, drawn uniformly at random from the complete set of 853, giving 19,900 pairs. The seven properties used as predictors are edge count, maximum degree, minimum degree, triangle count, average clustering coefficient, diameter and degree variance.

Graph edit distance is computed on a subset of sixty graphs, giving 1,770 pairs, because a full matrix at two hundred graphs was outside the compute budget. Its R^2 is therefore estimated from fewer observations than the other three, and it is additionally affected by the time budget discussed in Section 3.1.7. Both qualifications apply to the GED row throughout.

3.3.4 Procedure

The seven predictor columns are standardised so that coefficients are comparable. The response is standardised for the same reason. The full model is fitted by ordinary least squares, then seven reduced models are fitted, each omitting one predictor. A confidence interval on each unique variance is obtained by resampling pairs two hundred times and refitting.

3.3.5 Results

The four distances differ sharply in how much of them these seven properties account for.

Spectral is almost entirely elementary. Its R_{all}^2 is 0.954, leaving under five percent unaccounted for. Edge count explains 0.926 on its own and 0.169 uniquely,

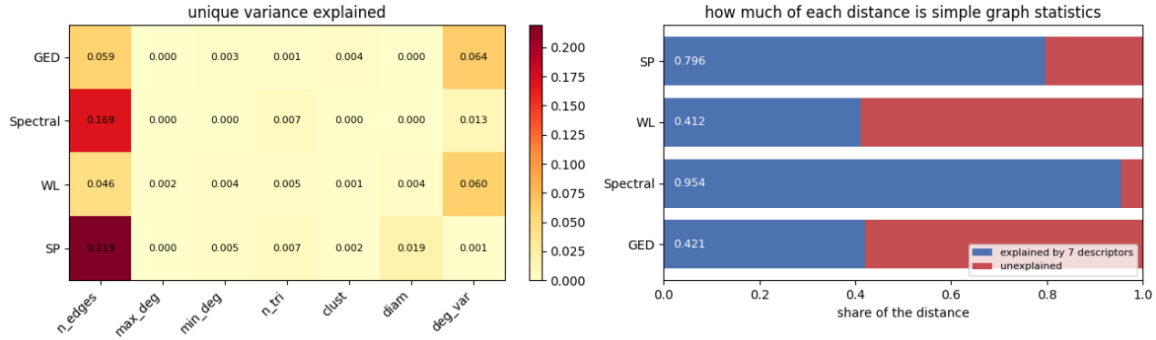


Figure 3.2: Left: unique variance per property and distance. Right: the share of each distance explained by the seven properties together, against the share unexplained.

the gap is shared with triangle count (0.701 marginal) and minimum degree (0.328 marginal), both of which rise with edge count. On this sample, the spectral distance behaves close to a reweighting of graph size.

SP is largely elementary, with a diameter component. Its R_{all}^2 is 0.796. Edge count again dominates, at 0.751 marginal and 0.219 unique. What distinguishes SP is diameter: its unique diameter contribution is 0.019, against 0.000 for Spectral and GED and 0.004 for WL. In absolute terms this is small, but it is the only non-negligible value in that column.

WL is the least accounted for. Its R_{all}^2 is 0.412, so roughly fifty-nine percent of its variation is not explained by any of the seven properties. Its unique contributions are small and spread across degree variance (0.060) and edge count (0.046) rather than concentrated anywhere.

GED sits between. Its R_{all}^2 is 0.421, so roughly fifty-eight percent is unaccounted for, close to WL. It differs from Spectral and SP in where its unique variance lies: degree variance is its largest term at 0.064, marginally above edge count at 0.059, whereas both spectral and shortest path distances are dominated by edge count alone. This row is the least reliable in the table, for the sample-size and time-budget reasons given above, a repeat run under a different time budget moved its R_{all}^2 from 0.462 to 0.421 while the other three rows were unchanged to three decimal places.

Figure 3.2 shows the unique variances as a heatmap and the explained share as stacked bars.

3.3.6 Interpretation

Four statements follow, each holding for connected graphs of order seven and for these seven properties.

The spectral distance is approximately a function of graph size on this population. Seven elementary numbers account for 95% of its variation, and edge count alone accounts for 93%. Anything the spectral distance appears to detect should therefore be checked against what edge count alone would detect, which is why the covariate baseline of Section 3.1.5 accompanies every later result.

SP is the only distance with an appreciable unique diameter component. This is a prediction rather than a conclusion, it says SP should be the distance that succeeds when diameter defines the classes, and Experiment 6 tests that. Note also that Experiment 1 found SP to be the most degenerate of the three, so a distance can discard a great deal and still retain one property sharply.

The Weisfeiler Leman distance measures something these properties do not capture. With 58.8% unexplained, most of what WL responds to is not any of the seven. The honest form of this claim is negative, it is not that WL is irreducible, only that it is not reducible to *these* descriptors. A property not included here might account for it.

Unique variance understates sensitivity when properties are correlated. A small unique contribution can mean either that a distance ignores the property or that the property never varies independently enough for the model to isolate it. On this sample edge count and triangle count move together, so neither can be credited separately. Distinguishing the two readings requires a design in which one property changes and the others do not, which is Experiment 3.

Two limitations bound the whole experiment. The model is linear in the property differences, and a distance responding non linearly would be understated. And the result is specific to the population sampled: on a collection of graphs with a different size distribution or a different density profile, the same distances could decompose differently.

3.4 Experiment 3: Response to a Single Controlled Change

3.4.1 Question

Experiment 2 ended with an ambiguity it could not resolve. When a property shows a small unique variance, two readings are possible: the distance may be blind to it, or the property may simply never vary independently enough for the model to isolate its effect. On a population of graphs almost everything moves together, so the second reading is always available.

This experiment removes the ambiguity by changing the question from correlation to response. Take one particular graph, change one property of it by exactly one unit, change nothing else, and ask whether the distance moves. A distance that cannot

tell such a change from a change of the same cost that alters nothing is blind to that property in the strictest sense the data allows.

3.4.2 Design

The mechanism is a *rewire*: remove one existing edge and add one edge that was absent. The number of nodes and the number of edges are identical before and after by construction. Since Experiment 2 showed edge count to be the dominant shared influence on every distance, removing it by construction rather than by matching is precisely what is required.

For each source graph G and each target property, two rewires are generated:

- H_{changed} , a rewire that moves the target property by exactly one unit,
- $H_{\text{unchanged}}$, a rewire that leaves the target property at its original value.

Both are one deletion plus one insertion, so under the unit cost model of Section 2.2 they cost the same to perform. The comparison is $d(G, H_{\text{changed}})$ against $d(G, H_{\text{unchanged}})$, paired on G so that variation between source graphs cancels (1).

The reported statistic is the effect size of the paired difference: the mean difference divided by the pooled standard deviation of the two sets of distances. A distance is recorded as responding to the property when the effect exceeds 0.2 and a bootstrap confidence interval on the mean difference excludes zero. Both thresholds are declared in advance.

This is the tightest confound control available in the thesis. It is not statistical adjustment and not distributional matching: the two graphs being compared differ by a single edit of identical cost.

3.4.3 Data

Forty source graphs of order eight with between ten and fourteen edges, drawn from the exhaustive pool. Four target properties are tested: diameter, maximum degree, minimum degree and triangle count. Rewires are constrained to preserve connectivity, so component count cannot be tested by this mechanism and is examined indirectly through minimum degree.

Order eight is used rather than nine because graph edit distance is better behaved there. Since the pairs differ by exactly two edit operations, the GED solver has a trivial search and its result is exact regardless of the time budget, so the GED row of this experiment does not carry the qualifications that apply elsewhere.

3.4.4 Procedure

For each target property, source graphs are drawn and rewires generated by repeated random attempts until both a changed and an unchanged rewire are found. Each of the four distances is then evaluated on both pairs. The bootstrap uses five hundred resamples of the paired differences.

target property	GED	Spectral	WL	SP
diameter	0.000	0.542	0.166	1.572
maximum degree	0.000	0.312	1.131	0.345
minimum degree	0.320	0.693	0.582	0.547
triangle count	0.224	−0.018	−0.072	0.289

Table 3.5: Effect size of a one-unit change in the target property, with edge count and node count held identical by construction. Larger values mean the distance responds more strongly to the change than to a cost-equivalent change that alters nothing.

target	distance	d changed	d unchanged	effect	95% CI	responds
diameter	GED	2.0000	2.0000	0.000	[0.000, 0.000]	no
diameter	Spectral	1.1676	0.9455	0.542	[0.071, 0.355]	yes
diameter	WL	0.6611	0.6460	0.166	[−0.015, 0.041]	no
diameter	SP	0.1320	0.0412	1.572	[0.068, 0.116]	yes
max. deg.	GED	2.0000	2.0000	0.000	[0.000, 0.000]	no
max. deg.	Spectral	0.9840	0.8552	0.312	[−0.064, 0.321]	no
max. deg.	WL	0.6954	0.6123	1.131	[0.060, 0.108]	yes
max. deg.	SP	0.0761	0.0545	0.345	[−0.001, 0.047]	no
triangles	GED	2.0000	1.9500	0.224	[0.000, 0.150]	no
triangles	Spectral	0.8799	0.8856	−0.018	[−0.135, 0.120]	no
triangles	WL	0.6158	0.6243	−0.072	[−0.050, 0.042]	no
triangles	SP	0.0822	0.0625	0.289	[−0.003, 0.043]	no
min. deg.	GED	2.0000	1.9000	0.320	[0.000, 0.250]	no
min. deg.	Spectral	1.0327	0.8171	0.693	[0.099, 0.340]	yes
min. deg.	WL	0.6634	0.5913	0.582	[0.033, 0.127]	yes
min. deg.	SP	0.1090	0.0650	0.547	[0.014, 0.073]	yes

Table 3.6: Mean distances to the two rewires, with the effect size and its bootstrap interval. A distance is recorded as responding when the effect exceeds 0.2 and the interval excludes zero.

3.4.5 Results

Graph edit distance returns the same value for every rewire. For diameter and maximum degree the mean distance is 2.0000 to both the changed and the unchanged rewire, and the effect is exactly zero with a degenerate confidence interval. This is not an approximation: a rewire is one deletion plus one insertion, so under unit costs it always costs two. The small non-zero values for triangles and minimum degree (1.95 and 1.90 against 2.00) arise where a rewire occasionally disconnects the graph and the connectivity guard produces a different edit, not from any structural sensitivity.

SP responds to diameter and to nothing else strongly. Its effect on diameter is 1.572, roughly three times Spectral’s 0.542 and an order above WL’s 0.166. On maximum degree it is 0.345 with an interval touching zero, so it is recorded as not responding.

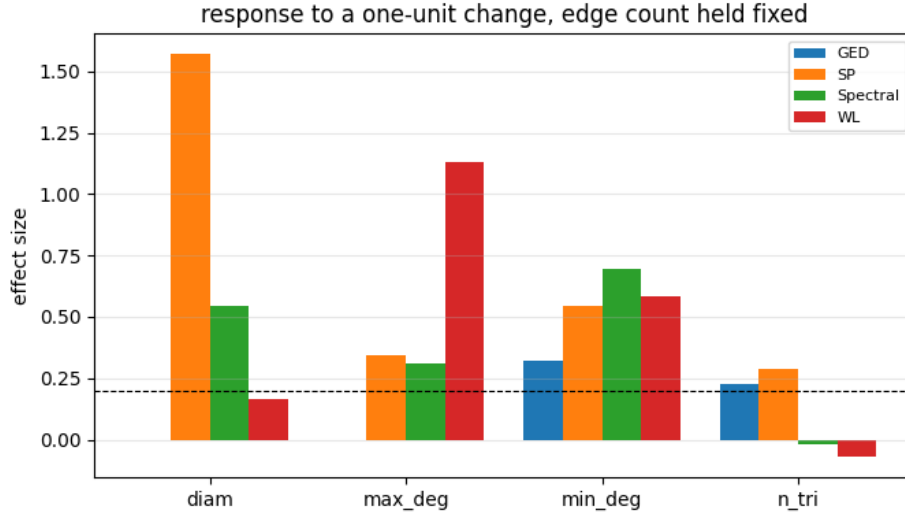


Figure 3.3: Effect size of a one-unit change in each property, by distance. The dashed line marks the declared detection threshold of 0.2, a bar above it still requires a confidence interval excluding zero to count as a response.

WL is the only distance that responds to maximum degree, at 1.131, while the other three do not.

No distance responds to a single triangle. With edge count held fixed the effects are 0.224, -0.018 , -0.072 and 0.289, and every interval includes zero. Two of the four are negative.

Minimum degree is the one property to which all three tractable distances respond.

Figure 3.3 shows the effect sizes against the detection threshold.

3.4.6 Interpretation

Four statements follow, each holding for connected graphs of order eight with ten to fourteen edges, for single unit changes, and for the unit cost edit model.

Under unit costs, graph edit distance measures the cost of a change rather than its consequence. Two rewires costing the same are at the same distance regardless of what they do to the graph. This is arithmetic rather than measurement, and it is the sharpest available characterisation of what this distance is. It is a statement about the cost model, not about edit distances in general, with weighted or structure dependent costs the behaviour would differ. The unit cost model is the one specified in the framework this work follows.

SP is specifically sensitive to diameter. It responds strongly to diameter and does not respond to maximum degree, which is what specificity means here. Taken with Experiment 2, where SP had the only appreciable unique diameter component, two independent methods agree.

An apparent sensitivity in Experiment 2 is resolved as confounding. Spectral showed a marginal triangle contribution of 0.701 there, but a unique contribution of only 0.007. This experiment now finds a pair level effect of -0.018 with an interval spanning zero. Three measurements agree that spectral distance does not respond to triangles once edge count is held fixed, and that its apparent association was carried entirely by edge count.

The triangle result generates a prediction. A graph is two colourable exactly when it contains no odd cycle, and the smallest odd cycle is a triangle. If no distance registers a triangle appearing or disappearing at fixed edge count, none should be able to separate graphs by chromatic number once edge count is matched. Experiment 9 tests this on data that plays no part in the present experiment.

Three limitations bound these results. The test uses single unit changes only, so a distance insensitive to one triangle might respond to ten, the claim concerns marginal response rather than sensitivity across the full range. Properties that a connectivity preserving rewire cannot alter, notably component count, are not tested directly and are approached through a proxy, which is an assumption rather than a measurement. And the effect sizes depend on the pool from which source graphs are drawn at a different order or density the same distances could respond differently.

3.5 Experiment 4: Class Separation and Its Relation to Recovery

3.5.1 Question

The three experiments so far describe the distances. The experiments that follow measure what clustering algorithms achieve on labelled datasets. This one sits between them, and exists because of a question that determines whether the later results can be read at all, before any algorithm runs, how far apart do the classes actually lie?

Without this measurement a low clustering score has three possible causes that cannot be told apart. The distance may be blind to the property, the dataset may have been built so that the classes barely separate, or the structure may have been present and the algorithm may have failed to find it. Measuring the middle case turns every later result into a deviation from an expectation rather than an isolated number.

The experiment therefore asks two things. What quantity best describes how separable a dataset is under a given distance? And what relationship does that quantity have with the recovery subsequently obtained?

3.5.2 Design

For every combination of dataset, ladder round, data seed and distance, three families of quantity are computed from the distance matrix alone.

Separation. The effect size d of Section 3.1.5, the separation AUC with its permutation null, and *min_gap*. The last is defined here, the distance matrix is embedded by classical multidimensional scaling, the mean position of each class along the first axis is computed, and *min_gap* is the smallest distance between adjacent class means, expressed in pooled within class standard deviations. It identifies which pair of neighbouring classes is the bottleneck, where d summarises the whole distribution.

Shape. *collinear* is the absolute rank correlation between class index and class mean position on the first axis, a value near one means the classes lie in order along a single direction. *n_dims* is the number of positive eigenvalues of the embedding, that is, how many dimensions the distance genuinely requires. *var_top2* is the share of positive eigenvalue mass in the first two dimensions, and *neg_mass* the share of total eigenvalue mass that is negative. A non-zero *neg_mass* means the distance is not embeddable in Euclidean space at all.

A reference clustering. One median SOM run on a 3×3 grid, merged to the true number of classes. This is not the clustering experiment, it exists so that the geometry can be calibrated against an outcome, and a single fixed configuration is used throughout so that the algorithm contributes no variation of its own.

Cells are then grouped into bands by separation and the mean recovery in each band reported. That table is the calibration against which the later experiments are read.

3.5.3 Data

All eleven dataset round combinations of Table 3.1, at four data seeds each, crossed with the distances available for that seed. Graph edit distance is computed on one seed per dataset for the reasons given in Section 3.1.7, the other three distances on all four. This gives 143 distance matrices in total, reduced to 44 cells after averaging over seeds.

3.5.4 Procedure

Each distance matrix is loaded from the cache, the three families of quantity computed, the permutation null obtained from three hundred label shuffles, and the reference SOM run at a fixed seed. Results are averaged over data seeds within each dataset round distance cell.

3.5.5 Results

Recovery rises monotonically across the four bands, from 0.030 below half a standard deviation to 0.760 above one and a half. Both ends of the range are well populated, with sixteen and fifteen cells respectively, so the relationship is not driven by a few extreme points.

Comparing candidate predictors:

separation band	mean ARI	std	cells
below 0.5 sd	0.030	0.059	16
0.5 to 1.0 sd	0.252	0.084	7
1.0 to 1.5 sd	0.365	0.145	6
above 1.5 sd	0.760	0.269	15

Table 3.7: Recovery as a function of achieved separation, measured before any clustering. Bands are in pooled within-class standard deviations.

distance	collinear	var_top2	neg_mass	n_dims
GED	0.955	0.440	0.234	23.4
Spectral	0.960	0.793	0.000	8.9
WL	0.886	0.308	0.000	43.7
SP	0.917	0.934	0.000	5.6

Table 3.8: Shape of the distance geometry. n_dims is the number of dimensions the distance requires, var_top2 is the share of that geometry a two dimensional plot can show, neg_mass above zero indicates a non-Euclidean distance.

statistic	Spearman correlation with ARI
separation AUC	+0.965
separation d	+0.957
min_gap	+0.877
var_top2	+0.331

Separation AUC and the effect size d predict almost equally well and both clearly outperform min_gap, which uses only the first embedding axis and therefore misses separation carried in other directions. The share of variance in the first two dimensions predicts poorly on its own.

The geometry of each distance, averaged over datasets:

SP requires about six dimensions, Spectral nine, GED 23 and WL 44. Correspondingly, a two dimensional plot shows 93% of SP’s geometry and 31% of WL’s.

GED has a negative eigenvalue mass of 0.234. The other three are exactly zero.

Collinearity lies between 0.886 and 0.960 for all four distances.

Figure 3.4 shows recovery against separation and the calibration by band.

3.5.6 Interpretation

Five statements follow, each holding for the datasets and distances used here and for the fixed reference configuration of the clustering step.

Recovery is closely governed by separation. Across 44 cells the rank correlation between achieved separation and recovery is +0.957, and the relationship is monotone across all four bands. On these datasets, knowing how far apart the classes lie before clustering is close to knowing the outcome.

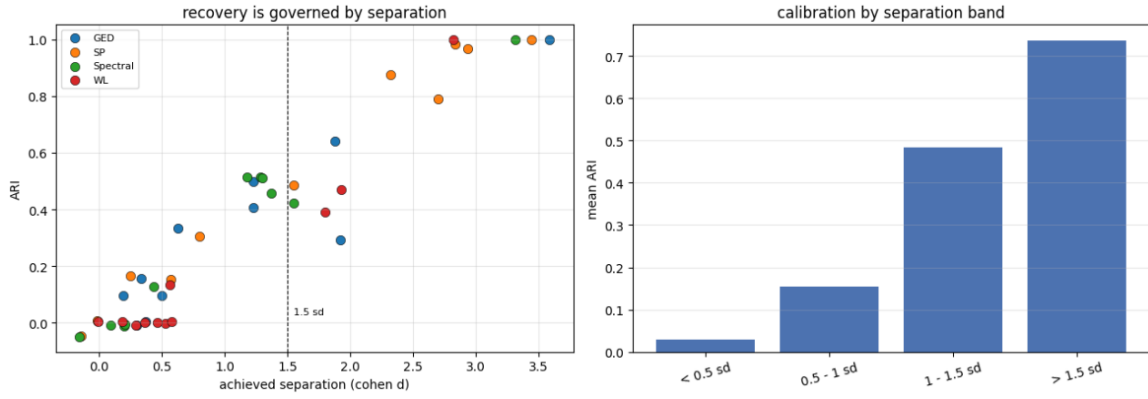


Figure 3.4: Left: recovery against achieved separation, one point per dataset round distance cell. The dashed line marks 1.5 standard deviations. Right: mean recovery within each separation band.

Reliable recovery begins near one and a half standard deviations. Below that the mean is 0.365 or lower, above it, 0.760. The threshold is a property of this setup rather than a universal constant, and it should be read as the point at which the present configuration begins to succeed rather than as a general law.

Separation AUC and the effect size are interchangeable as predictors, and the choice between them is practical. AUC correlates marginally better (+0.965 against +0.957), but a dataset cannot be designed toward an AUC, whereas it can be widened until its separation exceeds a stated number of standard deviations. The effect size is used as the design target throughout and the AUC as the reported statistic.

Graph edit distance is not Euclidean. Its negative eigenvalue mass is 0.234 here, and an independent run on different datasets at order seven gave 0.235. Since the quantity is stable across datasets and orders it appears to be a property of the distance rather than of any particular data. The practical consequence is that no flat plot of a GED matrix can faithfully represent it, which is the structural explanation for why two dimensional pictures of GED look worse than its separation numbers suggest.

The classes of every dataset lie in order along a single direction. With collinearity between 0.886 and 0.960, the three classes are three intervals of one continuous quantity rather than three separate clouds. Widening the class definitions, as was done for the diameter and edge count designs, moved the intervals further apart but did not remove the ordering. That the highest separation band nonetheless reaches an ARI of 0.760 indicates that an ordinal design is not by itself an obstacle, what matters is the size of the gaps rather than their arrangement.

Three limitations apply. The calibration is fitted with one clustering configuration on one family of designs, so the numerical threshold should not be transferred to other settings without recomputation. The reference clustering uses a single seed, so

seed to seed variation in the reference is not reflected in the band statistics. And `var_top2` correlates only weakly with recovery, so it should be read as a diagnostic for explaining individual cells rather than as a predictor.

3.6 Experiment 5: Comparing Clustering Algorithms

3.6.1 Question

Comparing clustering algorithms across datasets of unknown difficulty establishes little. A distance carrying almost no signal makes every algorithm look poor, a widely separated dataset makes every algorithm look good. When one method wins on one dataset and loses on another, the comparison cannot say whether that reflects the method or the problem.

This experiment removes the ambiguity by generating clustering problems across a controlled range of difficulty, measuring how difficult each one turned out to be, and plotting recovery against that measurement. What remains after difficulty is accounted for is attributable to the algorithm.

It also serves a second purpose. Every dataset so far has been built by selecting graphs on a property. The datasets here are built by a completely different process, so if the relationship between separation and recovery from Experiment 4 holds on both, it is not an artefact of either construction.

3.6.2 Design

The construction is a *planted partition*, three seed graphs are chosen. Each class consists of fifteen graphs obtained by applying r random edge rewirings to its seed, each rewiring preserving connectivity. The class labels are true by construction, since the seed each graph descends from is known.

Two parameters control difficulty. The perturbation radius r sets how far members drift from their seed, a small radius gives tight classes, a large one gives classes that overlap. The separation between the seeds sets how far apart the classes are placed to begin with.

Choosing seeds that are far apart requires a notion of distance, and fixing one would favour whichever distance was used. The seed rule is therefore swept rather than fixed. Four rules are used: uniform random selection, and maximin selection under the spectral distance, under the shortest path distance, and under a consensus formed by averaging the three normalised distance matrices. Reporting all four makes the dependence on this choice a measured quantity instead of a hidden assumption.

Difficulty is measured after generation rather than assumed. For each generated dataset and each distance the separation d is computed, and recovery is plotted against it. A distance that happens to place the seeds far apart simply occupies the easy end of the range for that distance, this is an observation about the distance, not a bias introduced by the design.

seed rule	radius	SP	Spectral	WL
random	1	0.76	1.13	0.76
SP	1	1.61	1.90	0.95
consensus	1	1.61	2.46	1.29
Spectral	1	1.11	2.85	1.42
random	8	0.03	0.05	0.04
Spectral	8	0.19	0.32	0.26

Table 3.9: Separation achieved under each seed rule, in within-class standard deviations, at the smallest and largest perturbation radii. No rule is neutral: maximin selection under the spectral distance gives that distance an easy problem and the shortest path distance a harder one, and the ordering reverses under the shortest path rule.

Four algorithms are compared, all producing three clusters so the comparison concerns which partition is formed rather than how many groups are returned, the median SOM on a 3×3 grid with prototypes merged to three, k -medoids, and agglomerative clustering under average and complete linkage (16).

3.6.3 Data

Seed graphs are drawn from the exhaustive pool of connected graphs of order nine with between ten and fourteen edges. Four seed rules are crossed with four perturbation radii ($r \in \{1, 2, 4, 8\}$) and three data seeds, giving 48 generated datasets of 45 graphs each.

Three distances are used: spectral, Weisfeiler Leman and shortest path. Graph edit distance is excluded because a single matrix at this size costs roughly few minutes and the full sweep was outside the compute budget. Results in this section therefore concern the three tractable distances only.

3.6.4 Procedure

For each generated dataset the distance matrix is computed, the separation d measured, and the four algorithms run. The SOM and k -medoids are run at three algorithm seeds and averaged, the two agglomerative methods are deterministic and run once. Cells are then grouped into separation bands and the mean recovery per band and algorithm reported. Variance in recovery is decomposed by band, seed rule, distance and algorithm.

3.6.5 Results

Random seed selection reaches only 1.13 standard deviations even at the smallest radius, so deliberate selection was necessary to generate easy problems at all. At the largest radius every rule produces separations near zero.

The recovery table shows small differences across algorithms within a band and large differences across bands. The variance decomposition quantifies this:

separation band	SOM	k -medoids	average linkage	complete linkage
below 0.5	0.024	0.055	0.026	0.040
0.5 to 1.0	0.123	0.246	0.118	0.237
1.0 to 1.5	0.363	0.420	0.448	0.440
1.5 to 2.5	0.585	0.547	0.665	0.716
above 2.5	0.921	0.554	0.943	0.943

Table 3.10: Recovery by difficulty band and algorithm. Reading across a row shows that the algorithms differ little at any fixed difficulty, reading down a column shows that difficulty changes the outcome greatly.

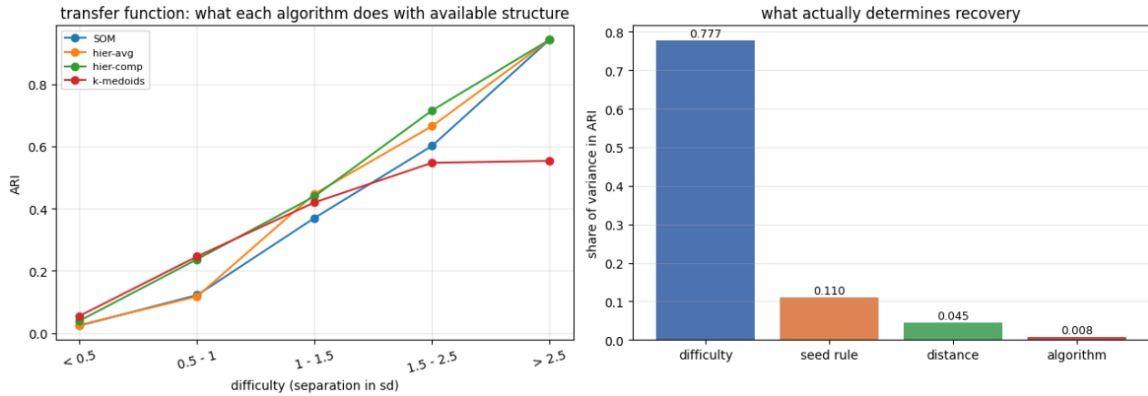


Figure 3.5: Left: recovery against difficulty for each algorithm. Right: share of variance in recovery attributable to each factor.

factor	share of variance in ARI
difficulty band	0.776
seed rule	0.106
distance	0.041
algorithm	0.009

Difficulty accounts for 77.6% of the variation in recovery. The algorithm accounts for 0.9%. The rule used to select seed graphs accounts for 10.6%.

At the easiest band the two agglomerative methods reach 0.943 and the SOM 0.921, while k -medoids stops at 0.554 and does not improve with further separation.

Figure 3.5 shows the transfer curves and the variance decomposition.

3.6.6 Interpretation

Four statements follow, each holding for these generated datasets, for graphs of order nine, and for the three tractable distances.

Difficulty dominates algorithm choice. On this data, how far apart the classes lie explains roughly 86 times more of the variation in recovery than which algorithm is used. The practical reading is that effort spent choosing between these four methods is small compared with effort spent on whether the representation separates the classes at all. This is measured on generated data with a fixed set of four algorithms and should not be extended to methods not tested here.

The relationship from Experiment 4 holds on a different construction. These datasets are made by perturbing seed graphs rather than by selecting on a property, and recovery still rises monotonically with measured separation. That the relationship survives a change of construction is evidence that it is not specific to either.

How the datasets are built matters more than which algorithm is run. The seed rule accounts for eleven times the variance that the algorithm does. Maximin selection under the spectral distance gives that distance 2.85 standard deviations and the shortest path distance 1.11, selection under the shortest path rule reverses the ordering. A study fixing one such rule without reporting it would be measuring its own construction. This is the reason the rule is swept here, and the same caution applies to any generated benchmark.

The algorithms differ chiefly at the easy end. Below 1.5 standard deviations all four lie within roughly 0.12 of one another. Above 2.5, agglomerative clustering reaches 0.943, the SOM 0.921, and k -medoids 0.554. On this data k -medoids does not benefit from additional separation beyond about 1.5 standard deviations, while the other three continue to improve. Why it plateaus is not established here.

Three limitations bound the experiment. Graph edit distance is absent, so no claim in this section covers it. Classes are generated by uniform random rewiring, which is not how categories of real graphs differ, so the transfer curves describe how these algorithms respond to this kind of structure. And the pool of order nine graphs proved tight, random seeds reach only 1.13 standard deviations, so the easy end of the range exists only because seeds were selected deliberately.

3.7 Experiment 6: Testing Predictions on the Designed Datasets

3.7.1 Question

The datasets of Section 3.1.4 are biased by construction. Choosing graphs by a property separates the classes on that property and on everything correlated with it, and Section 2.5 argued that this cannot be removed by building a cleverer dataset, matching runs out of graphs, and any selection rule is itself a graph property.

Such datasets can be used, provided they are used for a different purpose. A dataset built to demonstrate a claim proves little, because whoever built it knew what they wanted to see. A dataset used to check a prediction formed elsewhere is a genuine test, and its biases are tolerable because the prediction did not come from it.

This experiment asks whether the characterisations of Experiments 2 and 3, obtained without any designed dataset, predict what happens when clustering is actually run.

3.7.2 Design

Predictions are generated by a function that reads only the effect sizes of Experiment 3 and the list of covariates matched in each ladder round. It has no access to any clustering result. Writing it as a function rather than by hand has two consequences it cannot be adjusted after the outcome is seen, and it can be printed in full so the derivation is checkable.

The function assigns each dataset round distance cell one of three labels.

- If the target is the node labelling, the prediction is *succeeds* for the two distances that read labels and *blind* for the two that do not. This follows from the definitions rather than from any measurement.
- If the target is edge count, the prediction is *confounded*, since the target is itself the dominant covariate and no distance can be shown to be specific to it.
- If no covariate is matched in the round, the prediction is *confounded*, since edge count is free to vary and every distance responds to it.
- Otherwise the prediction is *blind* if Experiment 3 recorded no response to the relevant property, *succeeds* for the distance with the largest recorded effect, and *confounded* for the rest.

Chromatic number has no direct probe in Experiment 3, so triangle count is used as its proxy, this substitution is an assumption rather than a measurement.

Observed verdicts are assigned by the mechanical rule of Section 3.1.6: below the permutation null is *blind*, above the covariate baseline is *succeeds*, and in between is *confounded*. Scoring compares predicted against observed.

3.7.3 Data

All eleven dataset round combinations at four data seeds, crossed with the available distances, giving 44 cells after averaging. Predictions are recorded before the clustering is run.

3.7.4 Procedure

For each cell the separation AUC, its permutation null, and the best covariate baseline are computed, and a reference SOM clustering is run and merged to the true number of classes. The observed verdict follows from the first three, the ARI is reported alongside but plays no part in the verdict.

3.7.5 Results

Scoring the mechanical predictions against the observed verdicts over all 44 cells:

dataset	round	dist	AUC	null	baseline	by	ARI	observed
edit-wide	R1	Spectral	1.000	0.539	0.983	triangles	1.000	succeeds
edit-wide	R1	SP	0.989	0.536	0.983	triangles	1.000	succeeds
edit-wide	R1	GED	0.943	0.540	0.992	triangles	0.867	confounded
edit-wide	R1	WL	0.901	0.531	0.983	triangles	0.848	confounded
conn-wide	R3	SP	0.950	0.535	0.783	diameter	0.780	succeeds
conn-wide	R3	WL	0.611	0.531	0.783	diameter	0.079	confounded
conn-wide	R3	Spectral	0.560	0.524	0.783	diameter	0.015	confounded
conn-wide	R3	GED	0.551	0.527	0.757	diameter	0.048	confounded
geom-wide	R3	SP	0.972	0.539	0.895	max. deg.	0.889	succeeds
geom-wide	R3	Spectral	0.807	0.531	0.895	max. deg.	0.273	confounded
geom-wide	R3	GED	0.795	0.529	0.880	max. deg.	0.185	confounded
geom-wide	R3	WL	0.669	0.522	0.895	max. deg.	0.302	confounded
pure-label	R1	WL	0.994	0.545	0.500	—	1.000	succeeds
pure-label	R1	GED	0.916	0.536	0.500	—	0.423	succeeds
pure-label	R1	SP	0.460	0.537	0.500	—	−0.041	blind
pure-label	R1	Spectral	0.458	0.540	0.500	—	−0.048	blind
colorability	R3	GED	0.597	0.522	0.862	clustering	0.002	confounded
colorability	R3	Spectral	0.535	0.523	0.826	clustering	−0.011	confounded
colorability	R3	SP	0.503	0.526	0.826	clustering	−0.013	blind
colorability	R3	WL	0.503	0.521	0.826	clustering	−0.006	blind

Table 3.11: Selected cells from the ladder. Baseline is the best separation achievable by a single elementary graph property, named in the following column. The full table for all rounds is given in Appendix A.

predicted	observed		
	blind	confounded	succeeds
blind	4	8	0
confounded	0	18	8
succeeds	0	2	4

26 of 44 predictions are exactly correct, a rate of 59%. The off diagonal entries are all one step away from the diagonal, eight cells predicted blind were observed confounded, eight predicted confounded were observed to succeed, and two predicted to succeed were observed confounded. No cell predicted blind was observed to succeed, and none predicted to succeed was observed blind.

Two further observations from the ladder. On **conn-wide**, the spectral distance reaches 0.560 against a null of 0.524 at R3, with an ARI of 0.015, while the shortest path distance reaches 0.950 with an ARI of 0.780. On **geom-wide**, maximum degree remains unmatched at every round and attains 0.895 as a single covariate, the shortest path distance exceeds it at 0.972 while the other three do not.

Figure 3.6 shows the ladder collapse and the prediction outcomes.

3.7.6 Interpretation

Five statements follow, each holding for these datasets, these ladder rounds and the mechanical rules stated above.

Predictions from the distance-level experiments transfer, but only ordinally. The exact rate is 59%, which is modest. What is stronger is that no prediction inverted: in 44 cells the rule never predicted blindness where the distance succeeded, nor success where it was blind. Every error is a single step toward the

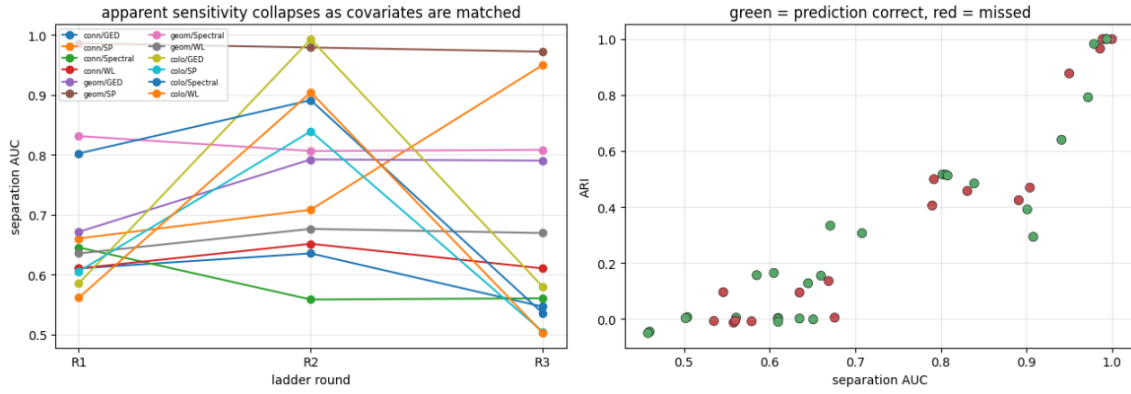


Figure 3.6: Left: separation AUC across ladder rounds. Right: recovery against separation, with cells coloured by whether the prediction was correct.

middle category. The characterisations therefore carry the ordering reliably while the exact boundary between categories does not transfer.

The errors are systematic and informative. Eight cells predicted blind were observed confounded, all at rounds where some covariate remained unmatched. Blindness to a property measured under exact control does not imply blindness on a dataset that still leaks a correlated covariate. Eight cells predicted confounded succeeded, all at unmatched rounds where the rule assumed that free edge count would prevent specificity two distances beat the covariate baseline regardless. Both errors reflect the rule being conservative rather than the characterisation being wrong.

The shortest path distance is specifically sensitive to diameter. It exceeds the best single covariate at 0.972 against 0.895 on **geom-wide** R3 with edge count and minimum degree matched. Maximum degree remains unmatched, so a residual confound is present, against this, Experiment 3 recorded an effect of 1.572 for diameter and 0.345 with an interval touching zero for maximum degree, which points to diameter as the operative property. The claim is supported by two independent measurements rather than established by either alone.

A property can be encoded in a representation and still be inaccessible to a distance built on it. The number of connected components is exactly the multiplicity of the eigenvalue zero in the Laplacian. Yet with edge count and both degree extremes matched, the spectral distance reaches 0.560 against a null of 0.524 and recovers almost nothing, while the shortest path distance reaches 0.950. On this dataset the information is present in the representation and the distance defined on it does not expose it. Whether this holds at other orders is not established, the same design at order seven gave a smaller gap.

The blindness control behaves as designed. On pure `label`, where the three classes share identical graph structures, the two distances that ignore node labels score 0.458 and 0.460 against nulls of 0.540 and 0.537, both below chance, with negative

ARI. This confirms both that the design isolates labelling and that the permutation null is calibrated correctly.

Three limitations apply. Several verdicts sit within 0.01 of a category boundary, the spectral distance on `colorability` R3 at 0.535 against a null of 0.523 is technically not blind but has an ARI of -0.011 , and the three way rule imposes a sharp cut on a continuous quantity. Maximum degree could not be matched on `geom-wide` at this order, so that dataset retains a residual confound. And the prediction rule uses triangle count as a proxy for chromatic number, which is an assumption carried from Section ?? rather than a measured relationship.

3.8 Experiment 7: Clustering Without Labels

3.8.1 Question

Every experiment so far has had an answer key. Datasets were built with known classes and recovery was measured against them. That is not how clustering is used in practice, if the labels were known there would be no reason to cluster. The intended application is exploratory, where a collection of graphs is given and the question is whether any grouping exists at all.

Two questions arise :

First, how does one know whether a clustering result is worth anything? Correctness cannot be measured, so some other criterion is needed.

Second, when a distance does group a collection of graphs, what is the grouping a grouping *of*? A partition is only useful if one can say what distinguishes its parts.

This experiment addresses both, first on graphs that were not designed at all, and then on the designed datasets with their labels hidden.

3.8.2 Design

Two criteria are used in place of recovery.

Repeatability. A result that changes when the algorithm is restarted is chasing noise rather than structure. The clustering is run five times from different random initialisations and the mean pairwise agreement between the resulting partitions recorded, using the adjusted Rand index to compare partitions with each other rather than with any ground truth. The same is done across six random 80% subsamples, comparing each subsample partition against the full data partition restricted to the same graphs. Structure that is genuinely present survives both, structure that is not, does not. This is the only notion of validity available without labels.

Characterisation. For the discovered partition, the proportion of variance it explains in each elementary graph property is computed. The property with the largest

population	dist	silh.	k found	util.	rep. seed	rep. sub	tracks	η^2
$n = 7$	GED	0.058	1.4	0.156	0.320	0.333	triangles	0.000
$n = 7$	Spectral	0.421	1.8	0.644	0.236	0.086	triangles	0.500
$n = 7$	WL	0.061	2.0	0.422	0.386	0.360	deg. var.	0.008
$n = 7$	SP	0.376	2.4	0.778	0.415	0.901	edges	0.402
$n = 8$	Spectral	0.373	2.0	0.400	0.417	0.602	edges	0.617
$n = 8$	WL	0.049	2.4	0.556	0.463	0.415	edges	0.418
$n = 8$	SP	0.434	3.4	0.556	0.452	0.491	edges	0.525

Table 3.12: Clustering an undesigned population. Tracks is the property the discovered partition explains best, and η^2 is the proportion of that property’s variance it explains. Repeatability is measured across algorithm seeds and across 80% subsamples.

value names what the partition has organised itself by. This uses no class labels, only the graphs themselves.

Two further quantities are recorded. The number of clusters is not fixed in advance, the prototypes are merged at every level and the level maximising silhouette is chosen, so the algorithm decides how many groups it has found. Utilisation, the proportion of neurons that win at least one graph, is reported because a map with few active neurons has not produced a usable partition regardless of any other statistic.

In the second part the designed datasets are used with labels withheld. Recovery is computed afterwards and reported as a diagnostic under the name `ari_check`. It never contributes to a verdict, it exists only so that the reader can see how the label free criteria relate to correctness in cases where correctness happens to be known.

3.8.3 Data

Part one uses 120 connected graphs of order seven and 120 of order eight, drawn uniformly from the exhaustive pools. There are no classes and no labels. Graph edit distance is computed on a 60 graph subset at order seven only, for cost reasons.

Part two uses the eleven dataset round combinations at one data seed each, with labels withheld during clustering.

3.8.4 Procedure

Clustering uses the median SOM on a 3×3 grid at five algorithm seeds. Prototypes are merged to the silhouette maximising level. Repeatability across seeds is the mean pairwise agreement over the ten pairs of runs, repeatability across subsamples is the mean over six 80% draws.

3.8.5 Results

On the undesigned population the three tractable distances produce different partitions of the same graphs. The spectral distance splits them by density, into a group averaging 10.3 edges and 3.8 triangles and a smaller group averaging 15.2 edges and

distance	group	size	edges	diameter	triangles
Spectral	0	103	10.33	2.83	3.80
Spectral	1	17	15.24	2.00	13.12
SP	0	95	11.82	2.49	6.09
SP	1	25	8.00	3.56	1.40
WL	0	51	11.14	2.76	5.08
WL	1	69	10.94	2.68	5.14

Table 3.13: Mean properties of the groups discovered in 120 undesigned graphs of order seven. The spectral distance separates sparse from dense, the shortest path distance separates compact from elongated, the two groups found by WL are indistinguishable on every property recorded.

13.1 triangles. The shortest path distance splits by shape, into groups averaging diameter 2.49 and 3.56. The two groups produced by WL differ by 0.2 edges and 0.08 in diameter, and its best explained property reaches $\eta^2 = 0.008$. Graph edit distance produces one group containing all sixty graphs.

Repeatability across algorithm seeds ranges from 0.236 to 0.463 over all seven population cells. Subsample repeatability is more variable, from 0.086 to 0.901.

On the designed datasets with labels withheld, four cells illustrate the range of outcomes:

dataset	dist	silh.	rep. seed	ari_check	tracks	η^2
pure-label	Spectral	0.571	0.720	-0.049	triangles	0.852
pure-label	WL	0.307	1.000	0.557	B labels	0.818
conn-wide R1	SP	0.485	1.000	0.076	diameter	0.430
geom-wide R3	SP	0.639	1.000	0.560	deg. var.	0.934

The `pure-label` dataset is built so that all three classes share identical graph structures and differ only in node labelling. The spectral distance ignores labels, so it cannot recover those classes, and its `ari_check` of -0.049 confirms that. It nonetheless produces a partition with silhouette 0.571 and seed repeatability 0.720, and that partition explains 85.2% of the variance in triangle count.

Figure 3.7 shows silhouette and repeatability against the withheld labels.

3.8.6 Interpretation

Four statements follow, each holding for these populations, for the median SOM on a 3×3 grid, and for the three tractable distances plus GED where computed.

Given the same graphs and no labels, the distances impose different organisations. On the undesigned population the spectral distance groups by density, the shortest path distance by shape, and WL by something none of the recorded properties explains. This is the clearest form in which the choice of distance can be seen as a modelling decision, nothing about the data or the algorithm differs between these three results, only the notion of similarity, and the notion of similarity determines what is found.

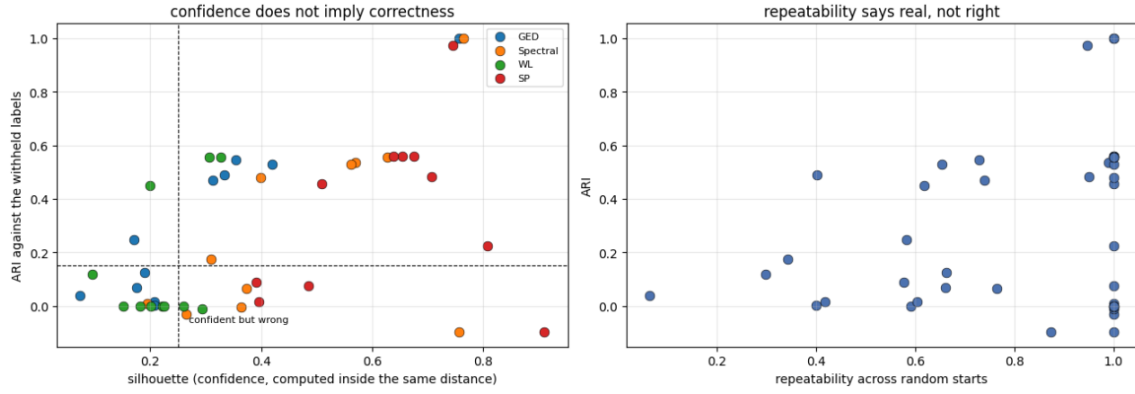


Figure 3.7: Left: silhouette against recovery on the designed datasets, with labels withheld during clustering. Right: seed repeatability against recovery.

High confidence in a partition does not imply that it is correct. On `pure-label` the spectral distance achieves a silhouette of 0.571 and perfect seed repeatability while recovering the intended classes at -0.049 , worse than chance. The dataset is constructed so that this distance provably cannot see the labelling, so the partition it found is necessarily something else, and the characterisation names it as triangle count at $\eta^2 = 0.852$. Internal confidence measures report that a partition is real, not that it is the one that was wanted.

On these undesigned populations the clusterings are not repeatable. Seed repeatability never exceeds 0.463, meaning the partition changes substantially between restarts. On this evidence, applying these distances to a broad sample of small graphs does not produce a stable grouping. The structure reported elsewhere in this chapter comes from datasets constructed to contain it. Whether the same holds for larger or more varied graph collections is not established here.

Without labels, repeatability and characterisation answer different questions, and neither answers the question of correctness. Repeatability indicates whether a partition is real, characterisation says what it is a partition of. Nothing available without labels indicates whether it is the partition one was looking for, and the `pure-label` result shows the two can diverge completely.

Three limitations apply. Repeatability establishes stability rather than usefulness: a partition could be perfectly reproducible and irrelevant to any question of interest. The characterisation checks only the nine properties recorded, so when WL’s split is explained by none of them the correct reading is “not these nine” rather than unexplainable. And the undesigned populations are of order seven and eight, where graphs are structurally crowded, poor repeatability at this scale does not establish that the same would occur among larger and more varied graphs.

3.9 Experiment 8: Anatomy of the Map and the Choice of Grid Size

3.9.1 Question

Experiment 5 found the self organising map competitive with hierarchical clustering but not superior to it, which raises a question for a thesis that takes the map as its primary method. If its partition is no better, what does it add?

The answer must lie in the arrangement. A partition says which graphs belong together, a map says which groups lie beside which others on a two dimensional grid, and how sharply they are divided. That arrangement is inspectable in a way a partition is not.

This experiment therefore asks three things. Is the arrangement meaningful, or is the grid position of a prototype arbitrary? Does the arrangement carry information about the quality of the clustering that the partition alone does not? And do the quantities conventionally used to judge a map support the use most often made of them, namely choosing the map's size?

3.9.2 Design

Four families of measurement are taken, all on the corrected median SOM described in Section 3.1.7.

Territories. Each neuron is assigned the majority true class of the graphs mapped to it, giving each class a territory on the grid. *Pieces* counts the grid-connected components a territory splits into and *concentration* gives the fraction of the territory lying in its largest component, both defined in Section 3.1.5. If the arrangement is meaningful a class should occupy few pieces with high concentration.

Arrangement against a null. Topographic error is reported both raw and as the ratio to the error obtained when the grid coordinates are randomly permuted. The ratio is the interpretable form, a value of one means the layout is no better organised than a random assignment of prototypes to grid positions.

Compression. Quantization error, the mean distance from a graph to its assigned prototype.

Grid selection. Three procedures are compared against fixed choices. The first takes the knee of the quantization error curve against K . The second takes the grid size maximising silhouette after merging. The third takes the knee of the silhouette curve. Each is scored two ways, how often it identifies the grid size that in fact gives the best recovery, and what mean recovery it delivers. The comparison baselines are the fixed choices $K = 4$ and $K = 9$, the latter being the conventional 3×3 default.

	pieces	concentration	TE	TE ratio	utilisation	ARI
class recovered ($\text{ARI} > 0.3$)	1.777	0.775	0.417	0.629	0.993	0.581
class not recovered	2.209	0.544	0.510	0.719	0.978	0.050

Table 3.14: Map properties where the clustering succeeds and where it does not. Territories are more concentrated and less fragmented when recovery succeeds, and the layout is better organised relative to a random one.

3.9.3 Data

The eleven dataset round combinations at one data seed, crossed with the available distances. Grid sizes $K \in \{4, 6, 9, 12, 16, 20, 25\}$ in approximately square shapes, with three algorithm seeds each. For the grid shape comparison, additional non square shapes are run at $K \in \{9, 12, 16\}$ with the neighbourhood width scaled to the grid diameter so that shape is not confounded with the amount of smoothing applied.

3.9.4 Procedure

For every combination the map is trained, prototypes merged to the true number of classes for the recovery measurement, and all four families of quantity recorded. The topographic null uses forty random permutations of the grid coordinates. Selection procedures are applied to the curves obtained by averaging over algorithm seeds within each dataset distance cell.

3.9.5 Results

Where recovery succeeds a class occupies 1.78 grid pieces on average with 77.5% of its territory in the largest piece, against 2.21 pieces and 54.4% where it does not. Utilisation is above 0.97 throughout, since the corrected map keeps every grid position active.

The topographic error ratio is 0.629 where recovery succeeds and 0.719 where it does not, with an overall mean of 0.676.

Correlating each map quantity with recovery across all runs:

quantity	Spearman correlation with ARI
concentration	+0.717
pieces	-0.144
topographic error	-0.193
topographic error ratio	-0.138
quantization error	+0.053

Turning to grid size, the curves behave as follows:

Quantization error decreases monotonically in K in 132 of 132 cells. Silhouette increases monotonically in 35% of cells. Every grid position remains active at every size.

The selection procedures compare as follows:

K	4	6	9	12	16	20	25
active neurons	4.0	6.0	9.0	12.0	16.0	20.0	25.0
quantization error	1.529	1.319	1.102	0.944	0.754	0.585	0.420
silhouette	0.371	0.391	0.409	0.423	0.426	0.428	0.434
clusters found	2.4	2.7	3.1	3.4	3.7	3.9	4.1
ARI	0.374	0.346	0.355	0.326	0.313	0.323	0.334

Table 3.15: Behaviour against grid size. Quantization error decreases monotonically and silhouette increases monotonically, while recovery is highest at the smallest grid tested and declines thereafter.

procedure	identifies the best K	mean ARI
knee of quantization error	11%	0.324
maximum of silhouette	34%	0.351
knee of silhouette	32%	0.353
fixed $K = 4$	48%	0.374
fixed $K = 9$	16%	0.355
best possible choice	100%	0.415

Table 3.16: Grid selection procedures against fixed choices, over 44 cells. “Identifies the best K ” counts cells where the chosen size is the size giving the highest recovery.

The two selection procedures agree with each other in 5% of cells. The knee of quantization error chooses $K = 9$ or $K = 12$ in 36 of 44 cells, the silhouette maximum chooses $K = 4$ in 15 cells and $K = 25$ in 9. The grid size that in fact gives the best recovery is $K = 4$ in 21 of 44 cells.

Grid size accounts for 0.3% of the variance in recovery. Which dataset and distance the cell concerns accounts for 95.6%.

For grid shape at fixed K , differences are small: at $K = 9$ the square grid gives 0.332 against 0.245 for the 1×9 line, while at $K = 16$ the 1×16 line gives 0.373 against 0.323 for the square. Shape accounts for roughly 3% of variance at fixed K .

Figure 3.8 shows the curves and the selection comparison, and Figure 3.9 shows one map in detail.

3.9.6 Interpretation

Five statements follow, each holding for the corrected median SOM, for these datasets, and for the grid sizes and shapes tested.

The arrangement carries information that the partition does not. Concentration correlates with recovery at +0.717, so how compactly a class occupies the grid is informative about whether the clustering succeeded. This quantity has no counterpart for k -medoids or hierarchical clustering, which return no arrangement. It is the clearest thing the map adds.

The layout is better organised than a random one, but not dramatically so. The topographic error ratio averages 0.676, so the observed arrangement makes

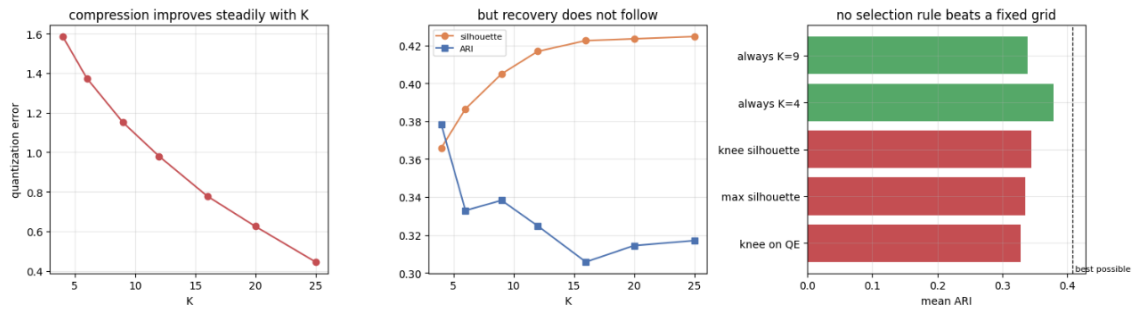


Figure 3.8: Left: quantization error against grid size. Centre: silhouette and recovery against grid size. Right: mean recovery delivered by each selection procedure, with the best possible choice marked.

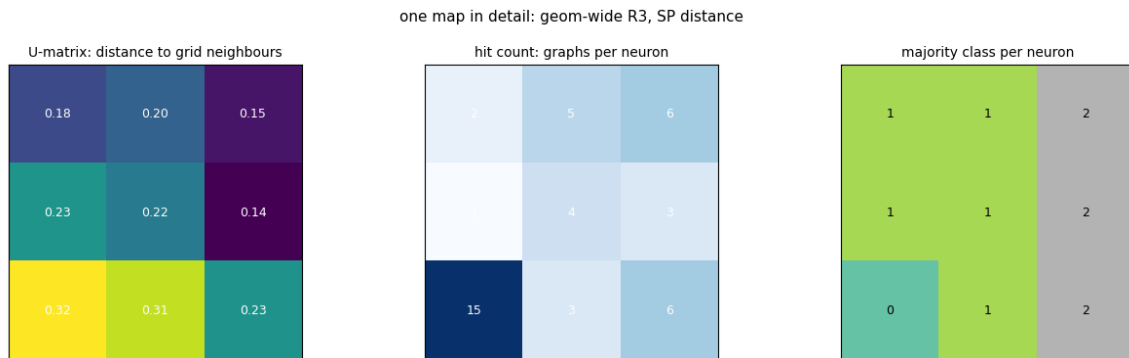


Figure 3.9: One map in detail: **geom-wide R3** under the shortest path distance. Left: the U-matrix, where light cells mark boundaries between dissimilar prototypes. Centre: the number of graphs assigned to each neuron. Right: the majority class of each neuron.

about two thirds of the errors that a randomly permuted grid would. This is a real effect and also a modest one. On this evidence the map's arrangement is meaningfully non-arbitrary.

Classes occupy few grid pieces but rarely exactly one. A recovered class splits into 1.78 pieces on average with 77.5% of its territory in the largest. The territories are therefore substantially but not perfectly contiguous, and the difference from unrecovered classes (2.21 pieces, 54.4%) is what makes concentration informative.

Quantization error is unrelated to recovery even though its curve is well behaved. It decreases monotonically in K in all 132 cells tested, which is exactly what the knee procedure assumes. Its correlation with recovery is +0.053. The failure of knee selection is therefore not caused by a malformed curve: the curve is correct and the quantity it measures is simply not the quantity of interest. Compression improves as prototypes are added, and recovery does not follow.

No selection procedure outperformed a fixed grid size on these datasets. The best procedure identified the optimal size in 34% of cells and delivered a mean

recovery of 0.353, against 48% and 0.374 for always choosing $K = 4$ and 16% and 0.355 for the conventional 3×3 default. The two procedures agree with each other in 5% of cells. Silhouette rises monotonically toward larger grids because larger grids permit tighter sub-clusters, and the number of clusters found rises from 2.4 to 4.1 across the sweep, so its maximum lies beyond the correct cluster count. The practical recommendation on this evidence is to fix the grid size in advance and demonstrate that conclusions do not depend on it, which is supported by grid size accounting for 0.3% of the variance in recovery against 95.6% for the dataset and distance.

Four limitations apply. That $K = 4$ is the best fixed choice was determined by inspecting these results, so the 48% figure benefits from hindsight, the conventional 3×3 default, chosen without hindsight, still outperforms both selection procedures on mean recovery. The comparison covers three procedures and does not exclude the possibility that some other criterion would succeed. Recovery is measured after merging prototypes to the true number of classes, so the grid size is not being asked to determine the number of clusters. And the grid sizes tested range from 4 to 25 on datasets of 33 to 60 graphs, so the largest grids approach one prototype for every two graphs.

3.10 Experiment 9: Case Study on Chromatic Number

3.10.1 Question

The properties used so far were chosen by the experimenter, and chosen partly because they could be controlled. This raises a reasonable objection: that the results describe the experimental setup rather than the distances.

This experiment addresses the objection by applying the same framework to a property that was not chosen for convenience. The chromatic number of a graph is of independent mathematical interest, computing it is NP HARD (6), and none of the four distances was designed with it in mind. If the characterisations of Experiments 2 and 3 are informative about the distances rather than about the designs, they should say something about this case too.

The question is whether any of the four distances can separate graphs by chromatic number once the properties that ordinarily accompany it are controlled.

3.10.2 Design

Chromatic number is difficult to isolate because the graphs that need many colours are usually also denser, more clustered and contain larger cliques. A single controlled dataset is not sufficient, so three are built, each removing one alternative explanation.

Round 1. Order nine with exactly fifteen edges in every graph, classes $\chi \in \{2, 3, 4\}$. Node count and edge count are fixed, so size cannot explain any separation. Clique

number is left free, a graph needing four colours typically contains four mutually adjacent nodes.

Round 2. Classes generated by Mycielski’s construction (26; 27) raises the chromatic number of a triangle free graph while keeping it triangle free. Cliques larger than an edge are therefore impossible in every class, removing the clique explanation.

Round 3. Order eight with edge count matched exactly across classes, contrasting $\chi = 2$ against $\chi = 3$. Size, node count and edge count are all controlled.

Alongside the separation achieved by each distance, two further quantities are reported. The covariate baseline gives the best separation obtainable from a single elementary graph property, answering whether the classes could have been separated by a trivial statistic. And η_χ^2 gives the proportion of variance in chromatic number explained by the partition a distance actually produces, which says whether the clustering has organised itself by the intended property or by something else.

3.10.3 Data

Round 1 uses sixty graphs, twenty per class. Round 2 uses thirty three graphs, produced by eleven distinct bipartite seeds each expanded twice. Round 3 uses fifty graphs, twenty-five per class. Each round is generated at four data seeds, graph edit distance is computed at one.

3.10.4 Procedure

For each round and distance the separation AUC, its permutation null, the covariate baseline, and the achieved separation d are computed. A median SOM is run and merged to the number of classes, giving recovery and the partition used for η_χ^2 and for the characterisation.

3.10.5 Results

At Round 1 the two strongest distances reach 0.854 and 0.829, while the best single property, triangle count, reaches 0.938. Every distance falls below that baseline. The partitions produced track the clustering coefficient rather than the chromatic number, although η_χ^2 is substantial for GED and Spectral at 0.733 and 0.723.

At Round 2 graph edit distance reaches 0.993 with an ARI of 1.000 and $\eta_\chi^2 = 1.000$: it recovers the classes exactly. The other three also exceed the covariate baseline. However, the classes of this round differ substantially in size: the per-class mean edge counts are 21.4, 27.1 and 56.3, giving the spread of 15.29 recorded in Table 3.1. The property the partitions track is maximum degree, not chromatic number.

At Round 3 all four distances fall to or below their permutation nulls. The largest AUC is 0.597 against a null of 0.522, the shortest path and Weisfeiler Leman distances reach 0.503 against nulls of 0.526 and 0.521, both at or below chance. Recovery is

round	dist	AUC	null	baseline	by	d	ARI	η_χ^2	tracks
R1	GED	0.854	0.526	0.934	triangles	1.550	0.476	0.733	clustering
R1	Spectral	0.829	0.524	0.938	triangles	1.317	0.490	0.723	clustering
R1	SP	0.631	0.518	0.938	triangles	0.359	0.112	0.052	clustering
R1	WL	0.585	0.518	0.938	triangles	0.275	0.072	0.180	deg. var.
R2	GED	0.993	0.550	0.849	max. deg.	3.625	1.000	1.000	max. deg.
R2	WL	0.904	0.541	0.847	max. deg.	1.925	0.375	0.458	max. deg.
R2	Spectral	0.891	0.538	0.847	max. deg.	1.553	0.316	0.562	max. deg.
R2	SP	0.839	0.546	0.847	max. deg.	1.552	0.448	0.511	diameter
R3	GED	0.597	0.522	0.862	clustering	0.351	0.002	0.024	min. deg.
R3	Spectral	0.535	0.523	0.826	clustering	0.095	-0.011	0.004	edges
R3	SP	0.503	0.526	0.826	clustering	-0.012	-0.013	0.006	diameter
R3	WL	0.503	0.521	0.826	clustering	-0.004	-0.006	0.004	triangles

Table 3.17: The three rounds. “Baseline” is the best separation from a single elementary property, named in the following column, η_χ^2 is the proportion of variance in chromatic number explained by the partition the distance produced, “tracks” names the property that partition explains best.

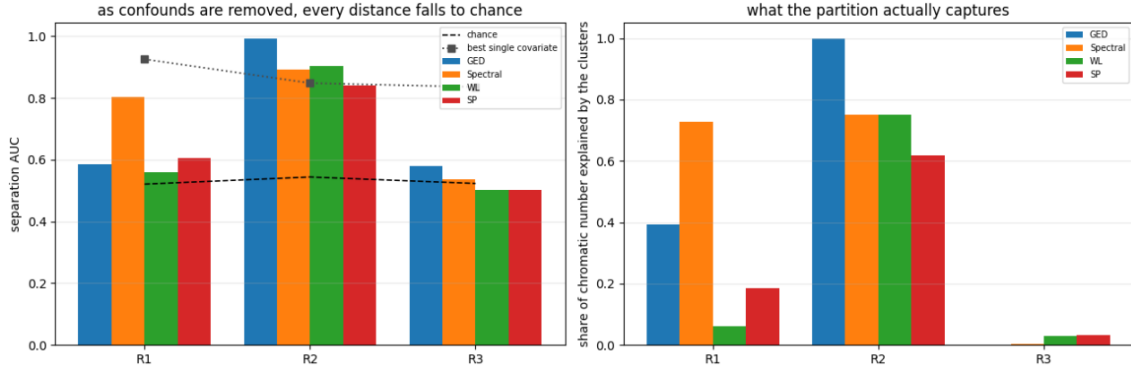


Figure 3.10: Left: separation achieved by each distance at each round, with the permutation null and the best single covariate marked. Right: the proportion of variance in chromatic number explained by each partition.

0.002, -0.011, -0.013 and -0.006. The variance in chromatic number explained by the partitions is 0.024 or less in every case. The best single property, average clustering coefficient, reaches 0.826.

Across the rounds the separation achieved changes as follows:

distance	Round 1	Round 3	change
GED	0.854	0.597	-0.257
Spectral	0.829	0.535	-0.294
SP	0.631	0.503	-0.128
WL	0.585	0.503	-0.082

Figure 3.10 shows the collapse across rounds against the null and the covariate baseline.

3.10.6 Interpretation

Four statements follow, holding for graphs of orders eight and nine, for the chromatic classes constructed here, and for the four distances as configured.

Round 2 cannot test chromatic sensitivity, and its result should not be read as one. Graph edit distance recovers the classes perfectly there, which would be a striking finding were it not that the three classes differ by a factor of nearly three in edge count, and that the partition tracks maximum degree at $\eta^2 = 1.000$. Mycielski's construction (26; 27) removes the clique confound at the cost of introducing a size confound, since each application roughly doubles the graph. The round demonstrates that controlling one property can install another, which is a methodological result, but it cannot support any claim about chromatic number. Round 3 is the only interpretable round.

Under edge count matching, no distance separates the chromatic classes. At Round 3 the four separations are 0.597, 0.535, 0.503 and 0.503 against nulls between 0.521 and 0.526. Recovery is at or below zero in every case, and the variance in chromatic number explained by the partitions does not exceed 0.024.

Remarkably, this property can be determined from a single elementary statistic. The average clustering coefficient separates the Round 3 classes at 0.826, well above every distance. This is the comparison that gives the result its force, had every method failed, the natural reading would be that the information is not present in graphs of this size. It is present, and easily extracted, and the graph distances do not use it. The mechanism is not mysterious (a two colourable graph is triangle free and so has clustering coefficient zero) and that specificity is also a limitation, since the same shortcut would not obviously separate $\chi = 3$ from $\chi = 4$.

The result was predicted by an experiment that used none of this data. A graph is two colourable exactly when it contains no odd cycle, and the smallest odd cycle is a triangle. Experiment 3 found that with edge count held fixed no distance responds to the appearance or disappearance of a single triangle, with effects of 0.224, -0.018 , -0.072 and 0.289 and every confidence interval spanning zero. The Round 3 outcome therefore follows from a property of the distances measured on rewired graphs of order eight, without reference to chromatic number at all. That a characterisation obtained without designed data predicts behaviour on designed data is the case the framework was built to make.

Four limitations apply. Round 3 contrasts only $\chi = 2$ against $\chi = 3$ on graphs of order eight, wider chromatic gaps or larger graphs may behave differently. The clustering coefficient baseline exploits the triangle free property of bipartite graphs and may not extend to higher chromatic contrasts. The graph edit distance figures carry the time budget qualification of Section 3.1.7. And Round 2 remains in the analysis as a methodological illustration only, it is not evidence about chromatic number in either direction.

Chapter 4

Discussion and Conclusion

4.1 Discussion

The experiments in this thesis were run separately, but they were never meant to be read separately. What follows draws them together. Everything stated here holds for the conditions we worked under (graphs of five to nine nodes, four distances, the datasets and algorithm settings described in Chapter 3) and we say so once rather than repeating it in every sentence. Where a claim rests on a single experiment we mark it as such.

4.1.1 What the four distances turned out to be

The four distances are usually presented as four ways of measuring the same thing. Our measurements suggest they are better understood as four different questions asked of a graph, and that the differences between them are larger than their definitions imply.

The clearest case is the spectral distance. Its definition promises access to global structure, and in one important respect it delivers exactly that, the number of connected components of a graph is precisely the number of zero eigenvalues of its Laplacian, so the information is present in the representation with mathematical certainty. Yet when we built a dataset whose classes differed in component count and matched edge count and both degree extremes across them, the spectral distance separated those classes at 0.560 against a chance level of 0.524, and recovered almost nothing. The shortest path distance, which has no such guarantee, reached 0.950 and recovered 0.780. Looking at what the spectral distance responds to across an undesigned population explains why: seven elementary graph properties account for 95.4% of its variation and edge count alone for 92.6%. The component information is there, but it is one small part of a spectrum dominated by size, and comparing whole spectra does not isolate it (14). A property can be encoded in a representation and remain unavailable to a distance defined on that representation. This is the single most useful idea to come out of the work, and it is not visible from any definition.

Graph edit distance produced a similar surprise in the opposite direction. It is the only distance that never confuses two different graphs, which is what makes it the

natural reference. But when we took a graph, moved one edge so that its diameter changed, and moved one edge so that nothing changed, it returned exactly 2.0000 in both cases. Under unit costs a rewiring is always one deletion and one insertion, so the distance measures what an edit costs rather than what it does. This is arithmetic, not a measurement, and it explains why an exact and expensive distance did not outperform cheaper ones in our clustering experiments. Expense buys completeness (no two distinct graphs are confused) but completeness is not the same as sensitivity to the property one cares about. Whether the same holds for weighted edit costs we cannot say, the unit cost model is the one specified in the framework we followed.

The shortest path distance is the mirror image. It discards more than any other distance we tested: on eight-node graphs it produces 198 different values for 11,117 graphs, and 99.6% of graphs share their value with at least one other. By any measure of information retention it is the worst of the four. Yet it was the most successful distance in our clustering experiments, because what little it keeps happens to include diameter, and it keeps that sharply. Three separate measurements agree on this: it has the only appreciable unique diameter component in the population regression, the strongest response to a single-unit diameter change, and it is the only distance to beat the best single-property baseline on our diameter dataset. Discarding information is not in itself a defect. What matters is whether what survives is what the task requires.

Weisfeiler Leman was the least reducible of the four. Seven elementary properties account for only 41.2% of it, it needs about forty-four dimensions to embed where the shortest path distance needs six, and when we clustered undesigned graphs with it the resulting groups differed by 0.2 edges and 0.08 in diameter, no property we recorded explained the split. We take this as evidence that it responds to something our descriptors do not name, and we are careful about how far that goes: the honest reading is “not these seven”, not “nothing describable”. It also did not translate into better clustering in our experiments. Retaining more structure and retaining the right structure are different things.

4.1.2 Should the distance be treated as a hyperparameter?

We think not, and the evidence is of two kinds.

The first is quantitative. On generated datasets where we controlled and then measured how hard each problem was, the difficulty of the problem accounted for 77.6% of the variation in recovery, while the choice among four clustering algorithms accounted for 0.9%. Since the difficulty of a problem is itself determined by which distance is used (the same graphs are easy under one distance and impossible under another) this says that the distance sets the ceiling and the algorithm merely approaches it. Tuning a hyperparameter usually means moving along a trade off, here the choice determines whether there is anything to find at all.

The second kind of evidence is more direct and, we think, more convincing. We clustered a hundred and twenty graphs drawn at random from the complete set of seven-node graphs, with no classes and no labels, using each distance in turn. The spectral distance split them by density, into a large sparse group averaging 10.3 edges and a small dense group averaging 15.2. The shortest path distance split the same graphs by shape, into a compact group of average diameter 2.49 and an elongated

group of average diameter 3.56. Weisfeiler Leman split them into two groups that no recorded property distinguishes. Nothing differs between these three results except the notion of similarity, and the notion of similarity determined what was found. That is what a modelling decision looks like: a commitment about what counts as meaningful, made before the analysis begins and not answerable to its outcome.

The practical consequence is that a distance should be chosen by asking what property matters and whether the distance responds to it, and this can be established before any clustering is run. Our prediction test suggests the characterisations transfer, though imperfectly: predictions generated from the distance-level experiments, without any access to the clustering results, were exactly right in 59% of cases and never inverted, no case predicted blind turned out to succeed, and none predicted to succeed turned out blind. The ordering carries, the exact boundaries do not.

4.1.3 What makes a graph clustering problem tractable

Across two quite different families of dataset (one built by selecting graphs on a property, one built by perturbing seed graphs) the single quantity that predicted recovery was how far apart the classes ended up. The rank correlation was $+0.957$ on the first family, and the same pattern held on the second, which is reassuring because the two constructions share nothing. Mean recovery rose from 0.030 where classes were less than half a standard deviation apart to 0.760 where they were more than one and a half apart. We would not transfer that threshold to another setting without recomputing it, but the shape of the relationship seems robust.

This has a consequence for how graph clustering studies should be designed. A dataset whose classes are less than about one standard deviation apart cannot distinguish between methods, because nothing recovers anything, a dataset whose classes are far apart cannot distinguish between them either, because everything succeeds. Useful comparison happens in a narrow band, and whether a dataset falls in it can be measured beforehand rather than discovered afterwards. We found this out the hard way: our first version of the diameter dataset used diameters two, three and four, and nothing recovered anything until we widened the classes to two, four and six.

Something else emerged from the same experiment that we did not expect. The rule we used to choose seed graphs when generating problems accounted for 10.6% of the variance in recovery, eleven times the algorithm effect. Choosing seeds to be maximally separated under the spectral distance gave that distance 2.85 standard deviations of separation and the shortest path distance 1.11, choosing under the shortest path rule reversed the ordering. This is why we reported all four rules rather than picking one. A benchmark generated with a fixed rule would favour whichever distance the rule was based on, and the effect is large enough to swamp the differences the benchmark was built to measure.

There is a final observation about the label-free case, which is where clustering is actually used. When we clustered graphs drawn at random rather than selected, no distance produced a partition that survived restarting the algorithm: agreement between runs ranged from 0.236 to 0.463 across every case we tested. All the structure reported elsewhere in this thesis comes from datasets built to contain it. We do not

know whether this reflects something about small graphs, which are structurally crowded, or something more general, but it is worth stating plainly: on an arbitrary collection of small graphs, these methods did not find stable groups.

4.1.4 What the self organising map is for

The map did not cluster better than the alternatives. At the easiest end of our difficulty range it reached 0.921 where average- and complete-linkage clustering both reached 0.943, and across the whole range the four algorithms we compared differed by very little. If the only output required is a partition, hierarchical clustering is simpler, deterministic and slightly better on our data.

What the map produces in addition is an arrangement, and the arrangement carries information. How compactly a class occupies the grid (concentration) correlated with recovery at +0.717. That means one can look at a trained map and form a view about whether the clustering worked, using a quantity that has no counterpart for k -medoids or hierarchical clustering, which return no layout at all. In the label-free setting this is worth something: the U-matrix and the territory structure give a reader something to inspect where a partition gives only a list.

We are careful not to overstate the arrangement's quality. Measured against a randomly permuted grid, our maps made about two thirds as many topographic errors (a real improvement, but not a large one) and a recovered class occupied 1.78 grid regions on average rather than one. The arrangement is meaningfully non-arbitrary and imperfect. We suspect this is partly a consequence of the median formulation, where a prototype must be an actual graph from the dataset and so cannot sit between two groups the way a constructed prototype could, but we did not test the constructive.

Our clearest advice about the map is negative and concerns its diagnostics. Quantization error falls as the grid grows (monotonically, in all 132 cases we tested, which is exactly what the standard elbow procedure assumes) and its correlation with recovery is +0.053. The curve is well formed and the quantity is simply unrelated to what one is selecting for. Silhouette rises steadily toward larger grids, because larger grids allow tighter sub-clusters, so its maximum sits past the correct number of clusters. Tested against fixed choices on 44 cases, the knee of quantization error identified the best grid size 11% of the time and the silhouette maximum 34%, against 48% for always using four neurons and 16% for the conventional 3×3 default, on mean recovery the fixed choices won as well. The two procedures agreed with each other in 5% of cases. We recommend fixing the grid in advance and demonstrating that conclusions do not depend on it, which is easy to justify given that grid size accounted for 0.3% of the variance in recovery against 95.6% for which dataset and distance was in use.

A related caution applies to internal validity measures generally. On the dataset where all three classes shared identical graph structures and differed only in node labels, the spectral distance (which ignores labels and therefore cannot possibly see those classes) produced a partition with silhouette 0.571 and perfect agreement across restarts, while recovering the intended classes at -0.049. That partition explained 85.2% of the variance in triangle count. The measure was not wrong: the partition it found was real, stable and compact. It simply was not the partition anyone was

looking for, and no quantity computed inside the same distance could have revealed that.

4.1.5 Whether expensive distances are worth their cost

Graph edit distance costs too much compared to alternatives at our graph sizes (few minutes for a single forty five graph matrix against under a second) and it did not repay that in our experiments. It never beat the best single elementary property on any of our matched datasets. On the undesigned population it produced a single cluster containing every graph. Its one clear advantage, that it never confuses two distinct graphs, did not translate into recovery, for the reason given earlier: it measures what an edit costs rather than what an edit does.

This should not be read as a general verdict. Our GED figures are computed with a time budget of 0.15 seconds per pair, and at order nine the solver takes too long. So we are not measuring graph edit distance so much as a bounded approximation to it, and the bound is loose. Our results at orders seven and eight, where the solver behaves better, are more trustworthy. Based on our experiments within a fixed computational budget, on graphs of these sizes, the expensive distance did not outperform the cheap ones, and there was no relationship between how much a distance retained and how well it clustered.

That last point is perhaps the most useful summary of the work. The shortest path distance discarded the most and clustered the best, graph edit distance discarded nothing and clustered poorly, Weisfeiler Leman retained the most structure that we could not name and performed in between. What predicted success was not how much a distance kept, but whether what it kept was what the task required, and that can be established before any clustering is run.

4.2 Conclusion

This thesis began with a question that turned out to be badly posed. Asking which graph distance is best for clustering invites an experiment whose answer cannot be attributed: the score reflects the distance, the dataset and the algorithm at once, and running the comparison more carefully does not separate them. Measuring the three apart, and using designed datasets only to test predictions formed without them, gave answers that a direct comparison could not have produced.

The picture that emerged is that a graph distance is best understood as a commitment about what counts as similar, made before the analysis and not answerable to its outcome. Which commitment one makes determines what can be found, the algorithm that follows determines only how much of it is actually recovered. On our data the first mattered far more than the second, and the property one is looking for can be present in a representation and still be unreachable by a distance built on it.

The self organising map came out of this in a mixed position. As a clusterer it was competitive but not superior, and the quantities conventionally used to tune it turned out not to support that use. As a way of laying out a collection of graphs so that the result can be looked at rather than merely reported, it offered something

the simpler methods do not. Whether that is worth its extra complexity depends on whether one wants a partition or an object to inspect.

All of this holds for small graphs and for the choices set out in Chapter 3. The natural next question is which parts survive outside those conditions, and the following two sections set out what stands in the way and how we would proceed.

4.3 Limitations

The most important limitation is graph size. Everything here concerns graphs of five to nine nodes, because exhaustive enumeration is what makes the indistinguishability counts exact and the covariate matching verifiable, and it stops being practical beyond nine. Small graphs are crowded: there are not many of them and they resemble one another. Several of our findings could be consequences of that crowding rather than properties of the distances, the severe degeneracy of the shortest path distance, the difficulty we had in generating well separated classes, and the failure to find repeatable structure in undesigned collections are all candidates.

The second concerns graph edit distance. Computing it exactly is NP hard, so we imposed a budget of 0.15 seconds per pair and took the best value found within it. At order nine this is not an approximation that converges: doubling the budget roughly doubles the computation and changes the answer, and two budgets a factor of ten apart gave distances correlated at only 0.60. Every GED result at that order is conditional on that choice, and we excluded GED from the algorithm comparison entirely for cost reasons. Our characterisation of it also depends on every edit operation costing the same, which is the model the framework we followed specifies but not the only one available.

Third, some confounds could not be removed. Matching discards graphs that cannot be paired, so each additional property matched shrinks the available pool, and on the diameter dataset matching maximum degree as well as edge count and minimum degree was infeasible at order nine. Maximum degree separates those classes at 0.895 on its own. The shortest path distance exceeds this and an independent experiment indicates it responds to diameter and not to maximum degree, but the confound is present and we report that claim as supported.

Fourth, our datasets are narrow in construction. Classes are defined by thresholds on a single property, so they lie in order along one direction in every case we measured, and classes formed by genuinely different generative processes might behave otherwise. The generated datasets used for the algorithm comparison are built by random rewiring, which is not how categories of real graphs arise. And our account of what a distance responds to is relative to seven elementary properties: when we say Weisfeiler Leman responds to something they do not capture, the correct reading is that our seven do not capture it.

Finally, a number of smaller choices could matter. The relationship between separation and recovery was calibrated with one clustering configuration at one grid size. The number of clusters is fixed to the true value before recovery is measured, so the grid is never asked to determine how many groups exist. The single-edit experiment tests changes of one unit, so a distance insensitive to one triangle might respond to

ten. And component count cannot be changed by a connectivity-preserving rewire, so it was approached through minimum degree as a proxy, which is an assumption rather than a measurement.

4.4 Future Work

The reasonable next step will be to push the pipeline to larger graphs, because graph size bounds every result in this thesis. Exhaustive enumeration is what makes the indistinguishability counts exact and the covariate matching verifiable, and it stops being practical past nine nodes. But not every experiment needs it. Counting how often a distance confuses two graphs works just as well on a large sample as on a complete set: the result becomes an estimate with an interval rather than an exact figure, which is enough to establish a trend. The shortest path distance went from confusing 71% of graphs at order five to 99.6% at order eight while Weisfeiler Leman stayed flat between 5% and 6%, and whether that divergence continues to orders fifteen or twenty is a concrete question with a cheap answer. The regression that characterises what a distance responds to also needs no enumeration, only a sample, so it transfers directly. What would have to be given up at larger sizes is the exact covariate matching, since drawing matched classes from a sample rather than a complete pool no longer guarantees that an infeasible design reflects a fact about graphs rather than an unlucky draw.

The second direction is to vary the cost model for graph edit distance. Our sharpest statement about that distance (that it measures what an edit costs rather than what an edit does) follows entirely from every operation costing one, which is the model our framework specifies. Assigning different costs to edge insertion, edge deletion and label substitution produces a family of distances rather than a single one, and the single-edit experiment of Chapter 3 can be repeated across that family unchanged. That would turn one observation into a map from cost vectors to structural sensitivity, and it would answer a practical question: whether an edit distance can be tuned to respond to a chosen property, and if so how much freedom the cost vector actually gives. This is inexpensive, because the experiment compares graphs differing by exactly two operations, where the solver is exact and the time budget never binds. It is also the one direction that could change our least flattering finding about the most principled distance.

Third, our account of what a distance responds to rests on seven elementary properties, and the largest unexplained share belongs to Weisfeiler Leman at 58.8%. We were careful to say that this means “not these seven” rather than “nothing describable”, but that caution is only honest if someone eventually tries more. Counts of small subgraphs, spectral moments of the normalised Laplacian, orbit counts and measures of local irregularity would all be candidates. Widening the battery improves two things at once. It sharpens the regression, turning a weak negative claim into either a positive identification or a stronger negative one. And it strengthens every confound control downstream, because the covariate baseline that accompanies each result is only as demanding as the list of properties it draws on: a distance that beats seven elementary baselines has cleared a lower bar than one that beats twenty.

Fourth, the confound ladder could be extended in the same spirit. We matched at most three properties simultaneously, and the ladder stopped where the pool ran out. With a larger pool (either from a bigger enumeration or from a sample) more properties could be matched at once, and the rungs could be built to remove the specific confound each result is exposed to. The clearest case is the diameter dataset, where maximum degree separates the classes at 0.895 on its own and could not be matched at order nine. Building that rung at a larger order would settle whether the shortest path distance's advantage there is diameter sensitivity or maximum-degree sensitivity, which is currently supported by an independent experiment rather than by the design itself.

Finally, the whole pipeline should be applied to a real collection of graphs. Everything here is synthetic, which is what made the controls possible, and the natural test of a characterisation is whether it predicts behaviour outside the setting it was measured in. The distance level experiments need no labels at all, so they transfer to any collection immediately. Where the collection carries labels, the prediction test applies unchanged: the characterisation is computed on the collection itself with the labels withheld, predictions are recorded, and only then is clustering run and scored. That is the same protocol used in Chapter 3, with a real population in place of an enumerated one.

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